

INTEGRATIVE CONCEPTUAL REVIEW

Leadership Emergence: An Integrative Review

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Despite significant scholarly attention and practical importance regarding who emerges as informal and formal leaders in organizations, an integrative framework of the leadership emergence literature remains elusive. The presence of such a framework proves integral for the advancement of work in this area due to the complexity of the field, coupled with its sprawling nature across multiple disciplines (e.g., management, communication, education, economics). Accordingly, in this review, we utilize a database of 270 primary studies to put forth a distal–proximal framework of leadership emergence. In particular, we systematically review past research to answer four questions: (1) what do we know about the phenomenon of leadership emergence itself, (2) what are the antecedents of leadership emergence, (3) what outcomes are associated with leader emergence, and (4) what are the boundary conditions of leadership emergence? By introducing a conceptual framework for informal and formal emergent leadership, we highlight areas of research maturity and nascency and offer several recommendations for future work in this domain. Altogether, we highlight broad theoretical implications for the leadership, teams, and individual differences literature—and elaborate upon several benefits that an integrated framework of emergent leadership provides for organizations.

Keywords: leader emergence, leadership, prospective leader attributes, integrative conceptual review, distal–proximal framework

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Organizations have long emphasized the importance of leadership as is evidenced in the billions of dollars invested by organizations in leadership-related efforts each year (SHRM, 2017; Westfall, 2019). Notwithstanding these organizational efforts related to ensuring the development and retention of formal and informal leaders, organizations still struggle to identify leaders and there remains a relatively limited understanding of the factors through which individuals emerge as leaders in the first place. Although most scholars would likely agree that leadership emergence involves individuals gaining influence over their peers, this phenomenon has long been a black box and there remains a lack of consensus related to what constitutes leadership emergence and how the process unfolds.

Questions surrounding leadership emergence, while seemingly simple on the surface, have posed a complex challenge for researchers as they have worked to better understand the nomological network of leadership emergence and the process itself. One of the underlying challenges faced by scholars has been the process of defining leadership emergence. In particular, scholars have grappled with the nuance between formal and informal leadership emergence. Although some

researchers have used the terms synonymously (Judge et al., 2002, 2004), others have drawn a clear distinction between the two concepts (Barling & Weatherhead, 2016; Luria et al., 2014; Shaughnessy et al., 2017)—a decision that has introduced concerns of construct clarity in the domain.

In addition, the leadership emergence literature has unfolded across seven decades with research appearing in academic journals across multiple disciplines including (but not limited to): communication (Jiang et al., 2015), criminal justice (Bishopp, 2013), economics (Grunau & Pecoraro, 2017), education (Leeming, 2019), healthcare (Asiamah & Mensah, 2019), management (Hu et al., 2019), and psychology (Schaumburg & Flynn, 2012). Although it is clear that leader emergence research is multidisciplinary in nature, it is far from interdisciplinary—with many research endeavors published in management and applied psychology outlets ignoring highly relevant studies published in other disciplines. The siloed nature of this research coupled with the maturity of the field—compounds the likelihood that researchers may inadvertently posit more novelty to research questions than likely merited.

An extensive search of the literature performed for this article yields interesting insights on components of the nomological network of leadership emergence that have been studied to date. While on the one hand, there has been significant progress through the studying of a subset of antecedents of leadership emergence—there remains a limited understanding of the role of mediators and moderators in explaining *why* individuals emerge as leaders and *when* individuals emerge into leadership roles. Further, consequences of leadership emergence such as whether the people who emerge will actually be effective, and whether those who emerge are satisfied with the

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informal or formal leadership role, have largely been ignored in the literature (cf. Barling & Weatherhead, 2016; Lanaj & Hollenbeck, 2015). Altogether, across this rich history of research, academics have offered significant insight into leader emergence, but we still have substantial headway to make before we fully understand how this concept fits into the broader leadership phenomenon. Accordingly, in this integrative conceptual review, we provide an overview of the current state of leadership emergence research and an agenda for future research. In so doing, there are several contributions this research has beyond past reviews in the field.

First, this research maps out the conceptual space occupied by both formal and informal leadership emergence. In particular, we review formal and informal leader emergence, discuss how the two concepts are interrelated in some ways and unrelated in other instances, and attempt to more fully integrate these two distinct literatures by identifying nuanced similarities and differences in their nomological networks. Second, we delineate the antecedents, outcomes, mediators, and moderators of both formal and informal leader emergence. This step builds upon past reviews of the topic by offering comprehensive insight into the entire nomological network of leader emergence instead of a single or small subset of antecedents (which has been common practice in reviews to date; e.g., Badura et al., 2018; Grijalva et al., 2015). Further, this research incorporates multiple disciplines of research that can help inform our understanding of leadership emergence in the workplace. Finally, we use this review to offer suggestions of future programs of research for the leadership emergence literature. By highlighting what has been done we will point out gaps in our understanding of leadership emergence that would benefit from future theorizing and help provide a theoretical foundation upon which further empirical research on the topic of formal and informal leadership can flourish.

In sum, it is our goal to advance the leadership emergence literature by providing a strong understanding of where the literature is currently at and where future research efforts related to leadership emergence would be most fruitful. In the sections to follow, we first outline what formal and informal leadership emergence is, why a review is warranted, and the scope of this review. Following these sections, we introduce an integrated theoretical framework leveraging past conceptual and empirical research on the topic. Through this approach, we help to highlight research that has been conducted to date—highlighting both consensus and dissensus across empirical studies—and noting future directions along the way (summarized in Appendix A in Supplemental Material). We now turn to the important theoretical conundrum of understanding leadership emergence itself.

What Is Leadership Emergence?

Interestingly, many articles related to leadership emergence fail to clearly address what is meant by leadership emergence and there appears in many cases to be a lack of consensus regarding what the construct entails. Although conceptualizations of leader emergence often vary across researchers, as noted above, they generally involve an individual coming to be viewed as influential by others (Judge et al., 2002; Lanaj & Hollenbeck, 2015). More specifically, we suggest that leader emergence encompasses the phenomenon of an individual becoming a leader within their team, organization, or among others to whom the prospective leader's actions matter, regardless of the formality of the leadership role. Thus, in our review, we draw from existing research to broadly conceptualize

leader emergence as the process through which an individual becomes influential to relevant others in a manner that involves the implicit or explicit granting of the leader role (DeRue & Ashford, 2010; Judge et al., 2002; Kaiser et al., 2008; Lanaj & Hollenbeck, 2015; Lee & Farh, 2019). Yet, we acknowledge that in some cases leader emergence can occur without *all* possible followers granting leadership to the focal individuals or perceiving them as especially leaderlike. Relatedly, we also suggest that leadership emergence will tend to involve leaders enacting the leadership role, although in some cases leadership emergence might occur without the focal individual internalizing or claiming the leader identity.

Because leader emergence research has frequently explored perceptions of leadership in the context of temporary, self-managed, or leaderless groups, some may assume that leader emergence only occurs in informal settings (Acton et al., 2019; Hanna et al., 2021). Notwithstanding, leader emergence also has the potential to occur in relation to more formal leadership roles, and scholars in various disciplines have specifically researched leader emergence in such contexts (e.g., Barling & Weatherhead, 2016; Li et al., 2018; Zhang & Arvey, 2009). Therefore, we suggest that informal leader emergence is manifested through the informal granting of the leader role by prospective followers in a manner that does not involve a change in prospective leaders' formal status or authority (DeRue & Ashford, 2010; Hanna et al., 2021), and that formal leader emergence tends to involve a more visible and clear granting of leadership from prospective followers or superiors that results in the prospective leader becoming a legitimately designated leader with formal status and/or authority over others. In the context of formal leadership emergence, the designation could be related to one's being granted an official role such as Chief Executive Officer (CEO), or it could simply be the result of others explicitly designating an individual as the formal leader of a group or other entity. Not surprisingly, formal and informal leadership emergence can be interrelated and reciprocal in nature. In sum, we argue that leader emergence is a dynamic process that is relevant to both informal and formal leadership roles, and will tend to involve perceptions, behaviors, and social interactions between the prospective leader and relevant others (e.g., prospective followers, superiors; DeRue & Ashford, 2010).

Why Is a Review Warranted?

The availability of meta-analyses and existing qualitative reviews on the topic of leader emergence raise the general question of why another review on the topic is necessary. We recognize the veracity of this concern and note two defining features of this review that depart from previous reviews on the topic. First, past reviews have focused on the concept of informal leader emergence largely staying silent on the role of formal leader emergence (e.g., Badura et al., 2018; Hanna et al., 2021; Mullen et al., 1989). As rightfully pointed out by a reviewer of this manuscript, "actual organizations entail webs of formal and informal leadership operating often in tandem." Accordingly, we bridge together these concepts shedding light on the convergence and divergence of findings to create greater consistency with how leader emergence operates in organizations. Second, past reviews on the topic have overwhelmingly focused on the factors giving rise to leader emergence (e.g., Ensari et al., 2011; Judge et al., 2002, 2004; Lord et al., 1986). As a result, questions remain regarding how the emergence process unfolds, under what contexts it may be impacted, and what are the relevant

consequences. Addressing these questions is a main aspect of this review. Guided by this focus, below we elaborate on the scope of this review in route to addressing these lingering questions within the leader emergence literature.

What Is the Scope of the Review?

As noted above, research on leadership emergence has been multidisciplinary appearing in domains such as management, psychology, criminal justice, communications, education, and health-care. For this reason, we initially conducted an expansive search for all peer-reviewed journal articles published before February 2020 using the PsycINFO and Web of Science databases.¹ Altogether, this original search yielded 69,367 articles. After removing non-English samples (5,426 articles), non-peer-reviewed work (e.g., newspaper articles and books; 28,667 articles), and duplicate articles across the databases (22,347 articles)—we were left with a total of 12,927 articles to evaluate for relevancy. At the request of the review team, we also performed a literature search for unpublished, peer-reviewed work using: (a) the ProQuest Dissertation and Theses Database and (b) the online programs for the Academy of Management Conference (1998–2020) and the Society for Industrial and Organizational Psychology Conference (2002–2020). Through this additional search, we identified 2,721 dissertations/theses for review, as well as 46 independent authors to contact regarding their unpublished conference submissions.

Each primary study was reviewed by a member of the research team to determine if informal or formal leadership was included as a part of the study. Of the 12,927 potential journal articles, 2,721 dissertations/theses, and 46 independent authors—we identified 270 primary studies that have empirically investigated informal and formal leadership emergence (the information extracted from each primary study was reviewed by two authors to ensure accuracy). Although we focus primarily on the quantitative research published to date, we nonetheless underscore the importance of theoretical and qualitative work that has laid a foundation for empirical work in this area including Lord and colleagues work on social information processing of leadership perceptions (Lord, 1985; Lord et al., 2001), as well as DeRue and Ashford's (2010) work that outlines the social construction process of leadership emergence.

Leveraging the aforementioned literature search, below we identify questions related to four core areas of research in the leadership emergence literature that allow us to evaluate the current state of this domain, while also enabling the establishment of an integrative framework to propel research on the topic into the future. In particular, we first review the literature unpacking the dynamic process of formal and informal leadership, and then delve into the research related to the antecedents, consequences, and moderators of formal/informal leadership emergence.

Similar to other integrative conceptual reviews, we focus our writing on the trends identified across studies which allow scholars in this area to take stock of where the literature is currently at and where it can move toward in the coming years. At the same time, in recognizing the merit of all scholarly work in the leadership emergence domain, and the value a summary provides to readers—we offer a detailed tables outlining the core findings of studies used for this review (see Tables 1–8). In this next section, we begin by reviewing the research related to the processes of both formal and informal leadership emergence.

Question 1: What Do We Know About the Phenomenon of Leadership Emergence Itself?

Because leader emergence research has long focused on what leads to emergence for formal and informal leaders (e.g., Bass, 1954; Mann, 1959; Stein, 1975), it would be logical to assume that we also have a robust understanding of the phenomenon of leader emergence itself. However, our review suggests that there has been limited research that has directly explored the social interactions and processes that constitute leader emergence. This is especially true when considering research of an empirical nature and research exploring the leader emergence phenomenon in formal contexts.

Although leadership phenomena have long been studied through a relational lens (e.g., Hollander, 1964; Katz & Kahn, 1978; Uhl-Bien, 2006), it has been more recently that scholars have begun to explicitly explore and unpack the black box of how these social processes are involved in leadership emergence (e.g., Day & Harrison, 2007; DeRue & Ashford, 2010; DeRue et al., 2009). This research has tended to be more theoretical in nature and uses a role identity perspective to unpack the phenomenon of how individuals become leaders. In addition, this research builds indirectly upon the important foundation of insights developed by previous scholars that sheds light on the social nature of this phenomenon (e.g., Kenny & Zaccaro, 1983; Lord et al., 1984; Murphy, 1941).

Research has tended to conceptualize leadership emergence as being inherently relational in nature because it involves interactive social processes and generally entails others acknowledging the prospective leader as a leader (Day & Harrison, 2007). Prospective leaders whose behaviors and traits align with the prospective followers' needs in that situation and implicit theories of a leader are likely to subsequently be granted the leader identity, and relational recognition and collective endorsement of the prospective leader, as a leader, occurs (DeRue & Ashford, 2010; Junker & Van Dick, 2014). This process has the potential to manifest in various ways depending on the context and individuals involved. Interestingly, although we suggest in our definition that granting is central to the leadership emergence process, some research in this domain also suggests the importance of the claiming of leadership by the prospective leader. For example, DeRue and Ashford (2010) theorized that the informal leadership emergence phenomenon revolves around followers granting the leader identity and the prospective leader claiming the leader identity, and that the reciprocal nature of the process is more important than the specific ordering of claiming and granting. They also suggest that through this process leader identities are established and reinforced, producing a leader-follower relationship that constitutes emergence. Though nuanced

¹ To ensure that studies of formal and informal leadership were included, we performed an abstract search using a comprehensive list of related terms, including: leader emergence, emergent leader, leader role occupancy, formal leader, informal leader, project leader, temporary leader, group leader, leader perceptions, leaderless group, shared leader, leader status, team leader, distributed leader, collective leader, peer leader, managerial status, and manager status. We also searched the above keywords replacing the word "leader" with "leadership." Finally, we paired the words promotion, claiming, and granting with leadership, leader, and manager. A similar search process was used for unpublished, peer-reviewed research (i.e., abstracts were searched for each of the above keywords, except: shared leader*, distributed leader*, collective leader*, and team leader*. These keywords were excluded as they did not produce unique studies in our original search of peer-reviewed work).

due to the formality of the granting of the leader identity, this theoretical understanding also has potential relevance to formal leadership emergence.

Research also suggests that the leadership emergence process involves prospective leaders consciously or unconsciously experimenting with a provisional self in the workplace or organization (Ibarra, 1999), that is then embraced and internalized as a formal or informal professional identity (DeRue et al., 2009). In addition, as prospective leaders emerge into formal and informal leadership roles there is also the possibility of identity conflict that necessitates role and identity crafting on the part of the prospective leader (Croft et al., 2015; Karelaiia & Guillén, 2014). Scholars have provided some empirical testing and support related to the relational nature of leadership identity construction through simulations (e.g., Marchiondo et al., 2015). This research suggests that the claiming and granting process is important from the perspective of observers in developing perceptions of leadership related to emergence. However, research has not theoretically unpacked or explored situations in which perceptions of the leaders' emergence are not shared by the prospective followers or leaders themselves. In addition, existing research has provided few insights into how the process unfolds psychologically for the participants involved in the granting and claiming process related to leadership emergence. This understanding is needed to unpack how many of the variables, and verbal and nonverbal behaviors that are related to leadership emergence, have their effect. It would also allow for a deeper understanding of how variables such as trust and group composition influence the psychological processes that occur (cf. Baker et al., 2016).

Though still acknowledging the social nature of leader emergence, other approaches to understanding this important phenomenon have been more leader centric and focused on the prospective leader making sense of and claiming the leader role. Using a narrative perspective, Zheng et al. (2021) introduced four frames (being, engaging, performing, accepting) that prospective leaders use as they navigate the emergence process. These frames allow the prospective leader to understand, articulate, and enact their identity as a leader (Zheng et al., 2021). The frames also facilitate prospective leaders in making sense of and engaging in identity work that allows them to construct a sense of coherence in their emergence as a leader. They also help prospective leaders to validate their identity claims as a leader. Finally, these frames help provide behavioral scripts as prospective leaders emerge as leaders and enact the role of leader. Though this approach is more leader centric, it still acknowledges the important role of followers in granting leadership and the relational nature of the emergence process.

Future Avenues for Research

There is still much empirical work that needs to be done notwithstanding the progress that has been made theoretically and empirically in understanding this important phenomenon. Acknowledging the potential challenges due to the complexity of leader emergence, additional research could test the validity of the relationships that were proposed by DeRue and Ashford (2010) and extend or challenge our current understanding. In addition to the qualitative approaches that have tended to be employed, survey and laboratory studies could explore some of the nuanced relationships and discrete interactions involved in leader emergence. Methods involving video that could capture the process would be especially interesting and

push forward the rigor of our understanding. Neuroscience might also be a complimentary approach to unlocking the complex neurological and psychological processes occurring for prospective leaders and followers related to emergent leadership (Waldman et al., 2011, 2017, 2019).

In addition, future empirical work that clarifies what granting and claiming in the leader emergence phenomenon tend to entail behaviorally would be useful. This research could unpack how these behaviors change and evolve during the various stages of leader emergence. It would especially be interesting to understand which claiming behaviors tend to be seen as overly aggressive or overly nuanced. Research might also consider how leader emergence differs based on context. For example, does leader emergence occur differently in organizations that are highly competitive, and ego driven, versus organizations that are more mission focused and collectively oriented. Relatedly, do personalized (self-focused) versus more socialized (other-focused) charismatic leaders tend to emerge in nuanced ways (Brown & Treviño, 2006)? Research considering how emergence might differ in various country cultures (e.g., high-power distance) and across country cultures (e.g., the prospective followers are from a low-power distance culture and the prospective leader is from a high-power distance culture) would also be useful and interesting.

Research could also explore how the emergence phenomenon is navigated differently by various personality types. For example, the phenomenon might be unique for a narcissist who is seeking to claim leadership through bold behaviors and dominance in comparison to a highly agreeable individual who may be prone to only claim leadership once it is granted from prospective followers. It could also uncover if there are unique stalling points during the leader emergence phenomenon for women and individuals from underrepresented minority groups that hinder their emergence. Such research might uncover inconsistencies in how prospective leaders from various demographic groups are assessed. Research could also consider how others' behaviors (e.g., acting as a surrogate or sponsor) might facilitate or undermine the leader emergence process for individuals from these groups (Galvin et al., 2010).

It is also important that research begins to more rigorously and specifically explore how the leader emergence phenomenon occurs in the context of formal leadership emergence and how it might be different than informal leader emergence. For example, are claiming-related behaviors less relevant in the formal context due to the explicit nature of the granting? The lack of research looking at the formal leader emergence phenomenon is likely in part due to leader emergence historically being largely considered in the context of leaderless groups. Research looking at formal leader emergence should differentiate between the more time-constrained selection processes involved with new hires and the longer-term emergence phenomenon involved with individuals with existing organizational membership (Carnes et al., 2015). In the context of existing employees emerging as formal leaders, it is likely that the process simultaneously involves informal leader emergence processes that facilitate the selection and designation of formal leadership. It is also likely that the process of formal leader emergence is less follower driven and more focused on how superiors or peers make sense of the prospective leader's potential and grant leadership. Many additional questions remain unexplored in this important area of leadership emergence research—and we underscore the necessity of building theory and conducting empirical work to

address these unanswered questions and advance the state of science in this area.

Question 2: What Are the Antecedents of Leadership Emergence?

Within the literature exploring leadership emergence, scholars have most frequently conducted research aimed at understanding the factors that contribute to the emergence of formal and informal leaders. Our review yielded that certain concepts are “regular suspects” in the literature (e.g., biological sex), appearing as the focal concept (or as a control variable) in a notable amount of studies—while other concepts have only been included in “one off tests” with little to no further exploration. Particularly relevant to the present integrative review is that research on the antecedents of leader emergence has often unfolded without the guise of an overarching theoretical framework—which leads to questions regarding where antecedents fit into the overall process of leadership emergence.

In our review, we combine DeRue et al.’s (2011) approach to understanding antecedents of leadership effectiveness with Kanfer’s (1990, 1992) approach to understanding distal and proximal factors that lead to work outcomes, to offer a useful organizing framework for understanding the antecedents and overall nomological network

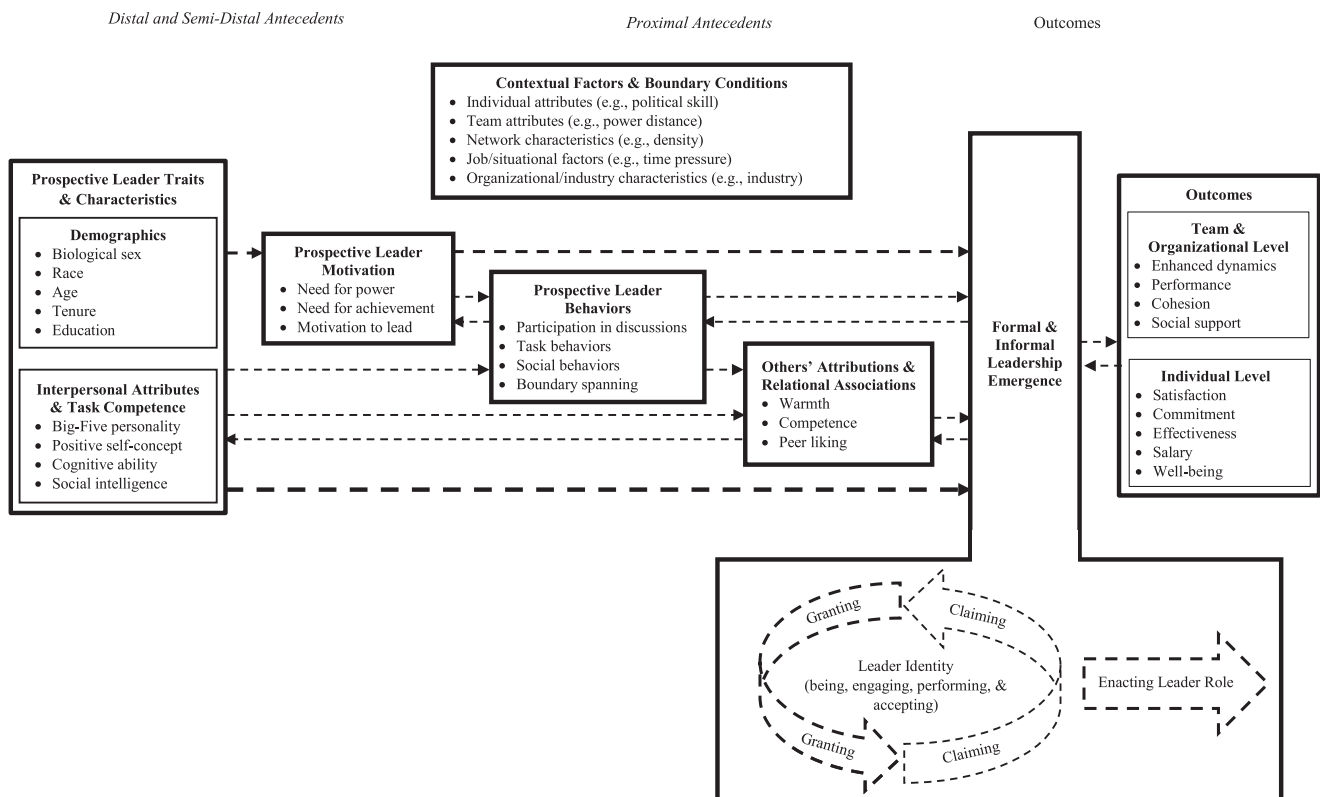
of leadership emergence (see Figure 1). This approach allows us to align the antecedents that have been researched theoretically based on their proximity to behavior—positioning surface-level characteristics such as demographic factors at the beginning of the process—and introducing more proximal factors such as prospective leader behaviors and attributions and relational associations into the theoretical model until the relevant outcome (in this case emerging as a leader) is achieved. We contend that much of the work that has unfolded on leader emergence can be organized using this approach—and doing so can highlight antecedents that have reached a state of research maturity—as well as nascent areas that require future inquiry.

Traits and Characteristics

Demographics

Some of the most distal antecedents from leadership emergence reflect demographic attributes that facilitate others’ perceptions that one can be characterized as a leader. Research on formal and informal leadership emergence has incorporated five such variables from this category with regularity including biological sex, race, age, tenure, and education (see Table 1). Notably, many of these

Figure 1
Integrative Framework of Leadership Emergence Research



Note. Depicted relationships and lists of variables are representative of reviewed research but not exhaustive in nature. Weight of the dashed lines roughly corresponds with the amount of existing research looking at a relationship or phenomenon, with lighter lines indicating nascent areas of research that might be especially fruitful topics for additional research. Details related to specific relationships and whether the relationships were explored in the context of formal and/or informal leadership emergence can be found in the manuscript and in Tables 1–8.

Table 1
Distal Demographic Antecedents of Formal and Informal Leader Emergence

Distal antecedents	Informal leadership emergence	Formal leadership emergence
Biological sex	[Favoring men]: Carbonell, 1984 (S1: .40*); Carbonell & Castro, 2008 (.37*); Dobbins et al., 1990 (N.R.); Flescher & Chertkoff, 1986 (S1: .36*; S2: .40*); Golub & Canty, 1982 (.33*); Hegstrom & Griffith (.33*); Hong et al., 2011 (S1: .12*); <i>Joe</i> , 2016 (.06*); Lanaj & Hollenbeck, 2015 (.17*); Lemoine et al., 2016 (S2: .09*); Malloy & Janowski, 1992 (.38*); Megargee, 1969 (S1: .70*; S2: .63*); Mitchell et al., 2019 (.12*); Neubert & Taggar, 2004 (.24*); Nyquist & Spence (S5*); Paunova, 2017 (.19*); Riggio et al., 2003 (.20*); Wentworth & Anderson, 1984 (.27*); [Favoring women]: Balthazard et al., 2009 (.21*); Chiu et al., 2021 (S1: -.15*); Gnagey, 1979 (-.11*); Kwok et al., 2018 (-.24*); McClean et al., 2018 (-.25*); Mitchell & Coyle, 2019 (.19*); <i>Park</i> , 2018 (S2: -.19*); Spillman et al., 1981 (-.25*); <i>Super et al.</i> , 2016 (-.19*). [Null]: Bergman et al., 2014 (.05 to .07); Brunel et al., 2008 (S1: -.04 to -.06; S2: .04 to .07); Carbonell, 1984 (S2: .19); <i>Chansler</i> , 2005 (.02); Chen et al., 2018 (.01); Chiu et al., 2017 (S1: .07 to -.01; S2: .16); Chiu et al., 2021 (S3: .02); Clark & Waldron, 2016 (N.R.); <i>Cohen & Yoon</i> , 2015 (.03); Colbert et al., 2012 (.02); Côté et al., 2010 (S1: -.04; S2: .05); <i>Dasborough et al.</i> , 2007 (-.10); <i>D'Innocenzo et al.</i> , 2014 (.11); Emery et al., 2011 (N.R.); Goktepe & Schneider, 1989 (.11); Gurman & Long, 1992 (-.01 to -.18); Ho et al., 2012 (-.14); Hong et al., 2011 (S2: -.08); Hu et al., 2019 (S1: -.13; S2: -.03); Kent & Moss, 1994 (.06 to -.09); Kolb, 1997 (.04 to .05); Kolb, 1999 (.10); <i>Lacerenza</i> , 2017 (S1 & S2: N.R.); <i>Lawson</i> , 2016 (-.02); Lee & Farh, 2019 (S1: .01 to -.02); Lemoine et al., 2016 (S1: .02); <i>Lim</i> , 2011 (-.17 to -.10); Lips & Keener, 2007 (.11); Lisak & Erez, 2015 (-.06); Liu et al., 2018 (-.05); Melwani et al., 2012 (S1b: .05; S3: .06); Mitchell et al., 2021 (S2: .02); Mitchell & Bommer, 2018 (-.09); Moss & Kent, 1996 (.05 to .02); <i>Mushman</i> , 1980 (-.04); Nevicka et al., 2011 (.12); <i>O'Rourke</i> , 1999 (-.08); <i>Park</i> , 2018 (S2: .15); <i>Roberts</i> , 1995 (.00 to .12); Schaumburg & Flynn, 2012 (.03); Schneider, 1978 (-.05); Schneider & Bartol, 1989 (-.08); Shaughnessy et al., 2017 (-.26); <i>Stigall</i> , 2005 (N.R.); <i>Trask</i> , 1999 (-.03); Türetgen et al., 2008 (-.07 to -.02); Walter et al., 2012 (S1: -.15; S2: .09); <i>Xu</i> , 2018 (.07 to -.13); Zhang et al., 2012 (.02). [Other]: <i>Cerny</i> , 2008 (-.20* to -.08); DeRue et al., 2015 (.04 to .06*); Geis et al., 1985 (N.R.); Hall et al., 1998 (.10 to .27*); Lee & Farh, 2019 (S2: .13 to .16*); Lim, 2019 (-.15* to -.10).	[Favoring men]: Bark et al., 2016 (.18*); Baxter & Wright, 2000 (N.R.); Li et al., 2011 (.05* to .13*); Ling et al., 2019 (N.R.); Moutafi et al., 2007 (.15*); Schuh et al., 2014 (S3: .30*); Schuh et al., 2014 (S4: .20*); Siegling et al., 2014 (.28*); Still, 1993 (N.R.); Tharenou, 2001 (.08* and .10*); Tharenou et al., 1994 (.16*); Wood, 2006 (N.R.). [Favoring women]: Barling & Weatherhead, 2016 (-.08*); Powell & Butterfield, 1994 (-.14*); Wyatt & Silvester, 2018 (-.34*). [Null]: Bagdadli et al., 2006 (N.R.); Bishopp, 2013 (N.R.); Breugh, 2011 (.02); Cannings & Montmarquette, 1991 (.18); Furnham et al., 2013 (.04 to .05); Li et al., 2018 (S3: -.01); Nelson et al., 2019 (N.R.); Salvador & Burciaga, 2020 (.08); Schuh et al., 2014 (S2: .18); Shaughnessy et al., 2017 (-.21); Weinberg et al., 2019 (.06). [Other]: <i>Middleton</i> , 2018 (.16 to .23*); Windsor & Auyeung, 2006 (N.R.)
Race	[Favoring White participants]: <i>Chansler</i> , 2005 (.39*); Charlier et al., 2016 (.31*); Chiu et al., 2021 (S3: .12*); [Favoring Non-White Participants]: Kwok et al., 2018 (.23*); [Null]: <i>Cohen & Yoon</i> , 2015 (.06 to .09); Mitchell & Coyle, 2019 (.09); Shaughnessy et al., 2017 (.23)	[Favoring White participants]: Barling & Weatherhead, 2016 (N.R.); Maume, 1999 (Male: N.R.); Nelson et al., 2019 (N.R.). [Null]: Bishopp, 2013 (N.R.); Maume, 1999 (Female: N.R.); Shaughnessy et al., 2017 (.17)
Age	[+]: <i>Chiu et al.</i> , 2014 (.15*) Elgar, 2016 (N.R.); Hong et al., 2011 (S2: .16*); <i>Joe</i> , 2016 (.05*); Lisak & Erez, 2015 (.13*); <i>Park</i> , 2018 (S1: .56*); <i>Trask</i> , 1999 (.26*); [-]: <i>Chansler</i> , 2005 (-.13*); Chaturvedi et al., 2012 (-.06*); DeRue et al., 2015 (-.17* to -.14*); Luria et al., 2019 (-.20*); Neubert & Taggar, 2004 (-.16*); <i>Zhang</i> , 2018 (-.13*). [Null]: Bass et al., 1953 (.13 to .20); <i>Bhattacharya</i> , 2019 (.03); Chiu et al., 2017 (S2: .04); <i>Dasborough et al.</i> , 2007 (.06); Hong et al., 2011 (S1: .05); Hunter & Jordan, 1939 (-.13); Liu et al., 2018 (.04); Melwani et al., 2012 (S3: -.11); <i>O'Rourke</i> , 1999 (.06); <i>Park</i> , 2018 (S2: -.07); Paunova, 2017 (.05); <i>Rutti</i> , 2009 (.15); Shaughnessy et al., 2017 (.17); Zhang et al., 2012 (.11). [Other]: <i>Karakowsky</i> , 1997 (.04 to .14*)	[+]: Bark et al., 2016 (.26*); Li et al., 2018 (S3: .11*); Maume, 1999 (Female: N.R.); Moutafi et al., 2007 (.27*); Schuh et al., 2014 (S3: .20*); S4: .29*); Tharenou, 2001; (.26* to .19*); Tharenou et al., 1994 (.39*); [-]: Asiamah & Mensah, 2019 (-.23*); Bishopp, 2013 (N.R.); Breugh, 2011 (-.19*); Furnham et al., 2013 (-.36* to -.25*); Salvador & Burciaga, 2020 (-.13*) [Null]: Bagdadli et al., 2006 (-.10); Barling & Weatherhead, 2016 (-.04); Maume, 1999 (Male: N.R.); Li et al., 2011 (.00 to .04); Ling et al., 2019 (N.R.); Opper et al., 2015 (N.R.); Schuh et al., 2014 (S2: -.07); Siegling et al., 2014 (.11); Shaughnessy et al., 2017 (.12); Windsor & Auyeung, 2006 (N.R.); Weinberg et al., 2019 (-.02). [Other]: Rosen et al., 1976 (-.11 to -.28*); Yap & Konrad, 2009 (Inverted U; N.R.)

Table 1 (continued)

Distal antecedents	Informal leadership emergence	Formal leadership emergence
Tenure/work/project experience	[+]: Chen et al., 2018 (.27*); Hong et al., 2011 (S1: .24*); Joe, 2016 (.12*); Kilduff et al., 2016 (.25*); Lawson, 2016 (.21*); Park, 2018 (S1: .67*); Zhang et al., 2012 (.17*), [–]: DeRue et al., 2015 (–.07*); [Null]: Chiu et al., 2017 (S2: –.03); Chiu et al., 2014 (–.13); Hong et al., 2011 (S2: .11); Hu et al., 2019 (S2: .08); Liu et al., 2018 (.04); Neubert & Taggar, 2004 (–.07); O’Rourke, 1999 (.15); Paunova, 2017 (.02); Zhang, 2018 (–.03)	[+]: Claussen et al., 2014 (.02* to .07*); Duncan, 2011 (.31*); Li et al., 2018 (S3: .22*); Ling et al., 2019 (N.R.); Nelson et al., 2019 (N.R.); Sieglung et al., 2014 (.21*); Tharenou, 2001 (.26* to .21*); Tharenou et al., 1994 (.20*); Windsor & Auyeung, 2006 (N.R.); [–]: Asiamah & Mensah, 2019 (–.30* to –.31*); Maume, 1999 (Female: N.R.); Rosen et al., 1976 (–.14* to –.31*); Wyatt & Silvester, 2018 (–.31*); [Null]: Bagdadli et al., 2006 (–.12); Borteyrou et al., 2015 (–.11 to –.06); Breugh, 2011 (–.12 to –.13); Kilduff et al., 2016 (.19); Maume, 1999 (Male: N.R.); Oppert et al., 2015 (N.R.); Powell & Butterfield, 1994 (–.08); Zhang & Arvey, 2009 (–.12); [Other]: Yap & Konrad, 2009 (Inverted U; N.R.); Weinberg et al., 2019 (–.05 to .11*)
Education level	[+]: Joe, 2016 (.32*); (2) Park, 2018 (S1: .26*); [–]: Wellman et al., 2019 (–.29*) [Null]: Bass et al., 1953 (.11 to .18); Colbert et al., 2012 (–.01); Hu et al., 2019 (S2: .10); Liu et al., 2018 (.00); Neubert & Taggar, 2004 (–.30); O’Rourke, 1999 (–.01); Zhang, 2018 (–.01); Zhang et al., 2012 (.04)	[+]: Li et al., 2011 (.09* to .13*; S3: .40*); Tharenou, 2001 (.07* to .08*); Tharenou et al., 1994 (.22*); Weinberg et al., 2019 (.14*); Yap & Konrad, 2009 (N.R.); Zhang & Arvey, 2009 (.21*); [–]: Asiamah & Mensah, 2019 (–.14*); [Null]: Bishopp, 2013 (N.R.); Ling et al., 2019 (N.R.); Maume, 1999 (N.R.); Oppert et al., 2015 (N.R.); Powell & Butterfield, 1994 (.02); Schuh et al., 2014 (S3: –.02; S4: .06); Windsor & Auyeung, 2006 (N.R.); [Other]: Grunau & Pecoraro, 2017 (N.R.)
Other distal attributes that have received limited research attention.	Attractiveness: [+]: Goktepe & Schneider, 1989 (.22*); Disability status: [–]: Luria et al., 2014 (–.16*); Height: [Null]: Elgar, 2016 (N.R.); Hunter & Jordan, 1939 (N.R.); Reynolds, 1944 (boys: .04; girls: –.07); [Other]: Page, 1935 (.01 to .12*); International student: [–]: Nahrgang, 2009 (–.46* to –.30*); Local language and English proficiency: [+]: Paunova, 2017 (Local: .24*; English: .19*); Manager/leader experience: [+]: Clark & Waldron, 2016 (.16*); <i>D’Innocenzo et al., 2014 (.57* to .62*); Edmunds, 1993 (.27*); Kilduff et al., 2016 (.23*);</i> Hong et al., 2011 (S2: .24*), [Null]: Bass et al., 1953 (.10 to .36*); Marital status: [Null]: Paunova, 2017 (.06); Nonnative English speaker: [–]: Hong et al., 2011 (S1: –.35*); Schaumburg & Flynn, 2012 (–.32*), [Null]: Hong et al., 2011 (S2: .08); Parent’s education: [+]: Hunter & Jordan, 1939 (N.R.); Political affiliation: [Null]: Hunter & Jordan, 1939 (N.R.); [Null]: Gruber et al., 2018 (S1: .04; S2: .12); Melwani et al., 2012 (S1b: .06); [Other]: Won, 2006 (.09 to .21*); Socioeconomic Status: [Null]: Bass et al., 1953 (–.04 to .03); Weight: [–]: Hunter & Jordan, 1939 (–.18*)	Attractiveness: [+]: Ling et al., 2019 (N.R.); Career interruptions: [–]: Yap & Konrad, 2009 (N.R.); Children: [+]: Tharenou, 2001 (.18* to .15*); [Null]: Maume, 1999 (N.R.); Windsor & Auyeung, 2006 (N.R.); Disability status: [–]: Luria et al., 2014 (–.09*); Family developmental experiences: [+]: Arvey et al., 2007 (.22*); Height: [+]: Melamed & Bozionelos, 1992 (Female: .30*; Male: .22*); [Other]: Li et al., 2011 (.05 to .11*); Location: [Null]: Zhang & Arvey, 2009 (–.08 to .05); Marital status: [+]: Tharenou, 2001 (.18* to .15*); [Null]: Maume, 1999 (N.R.); Windsor & Auyeung, 2006 (N.R.); Political affiliation: [+]: Ling et al., 2019 (N.R.); Prior influence experience: [+]: Atwater et al., 1999 (.18*); Oppert et al., 2015 (N.R.); Religion: [Higher for Christian Participants]: Asiamah & Mensah, 2019 (.12*); [Involvement]: Baker, 2003 (.51*); School quality: [+]: Barling & Weatherhead, 2016 (.08*); Socioeconomic Status: [+]: Barling & Weatherhead, 2016 (.06*); Li et al., 2011 (.06* to .10*); [Null]: Zhang et al., 2009 (.02)

Note. Italicized citations reflect unpublished, peer-reviewed studies. * denotes that effects reported in a primary study were statistically significant. N.R. indicates that there was insufficient information to calculate an effect.

attributes (e.g., biological sex) have tended to be incorporated into research as control variables when studying other phenomena of leadership emergence, nonetheless there are theoretical reasons which undergird the importance of these more surface-level attributes.

As an example, we turn to biological sex, which is the most studied demographic antecedent for both formal and informal leader emergence. Research on sex differences in leader emergence has often leveraged role congruity theory (Eagly & Karau, 2002), which argues that a misalignment between role expectations for women and leaders stifles the emergence of female leaders—a finding that has at times been supported in both the formal and informal leadership domains (e.g., Bark et al., 2016; Neubert & Taggar, 2004). At the same time, there has been research where no gender gap was present for formal and informal leader emergence (e.g., Lisak & Erez, 2015; Weinberg et al., 2019), and in a handful of instances, women were more likely to emerge as leaders (e.g., Barling & Weatherhead, 2016; Kwok et al., 2018). While the empirical findings of gender equality and a female advantage for leader emergence in certain situations may initially be surprising due to the often visible disparity of women's ascendance to high-level leader positions (e.g., women are less likely to occupy CEO positions)—variability in this effect has previously been demonstrated in meta-analytic research (Badura et al., 2018; Eagly & Karau, 1991) and explained via Eagly and Karau's (2002) role congruity theory (situations vary in the extent to which they create misalignment between women's and leader's role expectations).

Interestingly, a commensurate level of research has not yet been conducted related to race. Under the tenets of role congruity theory and other theories related to stereotypes such as expectancy violation theory (Burgoon, 1978; Burgoon & Jones, 1976) and the stereotype-content model (Fiske et al., 2002)—one would expect a leadership emergence disadvantage for members of racial minority groups. This finding has only been partially supported when studying formal leader emergence, as some studies show that racial minorities are less likely to emerge as leaders (e.g., Barling & Weatherhead, 2016), whereas other studies indicate that race does not serve as a significant predictor (e.g., Bishopp, 2013). Despite initial evidence related to formal leader emergence, this question has received inconclusive evidence in the context of informal leadership (e.g., Charlier et al., 2016; Kwok et al., 2018; Mitchell & Coyle, 2019), making it difficult for a conclusion related to this question to be validly drawn at this time. Thus, additional research is needed to understand the veracity of disadvantaged informal and formal emergence for members of racial minorities—with careful consideration of the intersection of race with other surface-level attributes such as gender and age.

Education has more frequently been incorporated into studies of formal rather than informal leadership. Findings indicate that elevated levels of education either favorably contribute to one's formal leadership emergence (e.g., Tharenou, 2001; Weinberg et al., 2019) or have no impact at all (e.g., Powell & Butterfield, 1994; Schuh et al., 2014). This conclusion is largely consistent with the idea that education increases one's collection of knowledge and skills (Tharenou et al., 1994), which can increase competence and confidence as it relates to performing the role of leader. Interestingly, education has been found to be generally unrelated to informal leadership emergence (e.g., Colbert et al., 2012; Zhang et al., 2012). Despite this trend, we caution readers that more research

is needed related to education and informal leadership as many studies of informal leadership reflect college students participating in lab studies or classroom projects rather than working adults (Badura et al., 2018). In other words, research needs to tease out whether informal leadership emergence is in fact unrelated to education or if it is a product of range restriction stemming from the types of samples examined.

Finally, tenure and age which are also often incorporated in research as control variables have showed a rich array of effects, leaving the status of these variables in question. For example, being older or working in a company for an extensive period of time reflects an impediment to emerging as a formal or informal leader (e.g., Asiamah & Mensah, 2019; DeRue et al., 2015), a finding that may be related to ageism stereotypes (e.g., the belief that older workers are less mentally agile or motivated to advance; Posthuma & Campion, 2009). In contrast, it has also served as a benefit (e.g., Lisak & Erez, 2015; Tharenou et al., 1994) or been deemed as irrelevant (e.g., Bagdadli et al., 2006; Chiu et al., 2017), perhaps due to the experience and knowledge one gains as they age or become more tenured. Importantly, research that has incorporated age and tenure may be underestimating the complexity of these demographic attributes as some initial evidence indicates the possibility of curvilinear effects on leadership emergence (Yap & Konrad, 2009). In other words, being too young or old; or lacking enough or too much tenure, each can prevent people from emerging into informal and formal leadership positions. Thus, moderate levels of age and tenure may facilitate one obtaining leadership positions—an assertion that would benefit from further theoretical and empirical investigation.

Interpersonal Attributes and Task Competence

Another set of commonly investigated distal attributes that have been studied in the formal and informal leadership emergence domains include the Big Five personality traits, cognitive ability, social intelligence (e.g., emotional intelligence, political skill), and positive self-concept traits (e.g., core self-evaluation, self-efficacy). We note that each of these attributes could serve as the beginning of a causal chain explaining who emerges as a leader or alternatively may serve as potential mediating mechanisms explaining the association between the distal antecedents mentioned in the prior section and leadership emergence.

Among the most commonly studied antecedents of leadership emergence are the Big Five personality traits. The first of these traits—extraversion—has often been classified as the quintessential personality predictor of leadership (Judge et al., 2002; cf. Grant et al., 2011). Surprisingly, our review of the literature indicated that the purported benefits of being extraverted for leader emergence is far less clear than originally expected. That is, while there is clear evidence that extraversion can aid an individual in becoming a formal leader (e.g., Arvey et al., 2006; Moutafi et al., 2007) or informal leader (e.g., Lee & Farh, 2019; Serban et al., 2015)—there are nearly as many instances where extraversion neither helps nor hurts one's likelihood of leader emergence (e.g., Byrne et al., 2007; McClean et al., 2018; Reichard et al., 2011). Collectively, these findings suggest that extraversion may not be as beneficial for leader emergence as once previously assumed—an important theoretical nuance to past meta-analytic conclusions that warrants additional scholarly attention. Agreeableness also exhibits a nuanced pattern of effects as it is largely unrelated to formal leadership emergence

(e.g., Byrne et al., 2007; Furnham et al., 2013)—yet, the impact on informal leadership is inconclusive as agreeableness has been shown to help (e.g., Walter et al., 2012), hurt (i.e., McClean et al., 2018), and be irrelevant (e.g., Lee & Farh, 2019) toward becoming an informal leader in a group. The remaining Big Five traits—emotional stability, conscientiousness, and openness to experience exhibit similarly beneficial or neutral implications for both formal and informal leader emergence (e.g., Colbert et al., 2012; Hu et al., 2019; Ogunfowora & Bourdage, 2014).

Positive self-concept traits including core self-evaluation, self-efficacy, and self-esteem also offer important insights into who emerges as leaders. Significantly more attention has been paid toward the role that positive self-concept has on informal than formal leadership emergence. Yet, the findings to date are remarkably similar indicating that viewing the self in a favorable light either helps informal or formal leader emergence (e.g., Atwater et al., 1999; Paunova, 2017) or bears no impact on emergent leadership (e.g., Kalish & Luria, 2016; Luria & Berson, 2013 [Study 2]).

Other attributes of a leader that are related to task competence are also relevant antecedents of leadership emergence. One such antecedent worthy of discussion is cognitive ability—which has long been viewed as a prototypical attribute of leaders that is relevant to leadership performance (Fiedler, 1995; Gordon, 1952; Lord et al., 1986; McCuen, 1929). Judge et al. (2004) underscored the linkage between intelligence and informal leader emergence, highlighting that “Lord et al., 1986 found that intelligence was the only attribute that is seen as a critical feature that must be possessed by all leaders” (p. 542) and indicated that the relationship “may be viewed by some as common sense” (p. 547). Despite the expected linkage between intelligence and emergent leadership, in reality the findings have been more mixed. That is, a relatively large number of studies have indicated that intelligence can be beneficial (25 studies; e.g., Li et al., 2011; Rubin et al., 2002) or irrelevant (19 studies; e.g., Atwater et al., 1999; McClean et al., 2018) to one’s ascendance into formal and informal leader positions. Accordingly, there remains an unexplored paradox that intelligence has the potential to both enhance and be irrelevant to one’s chances of emerging as a leader in an organization.

Beyond cognitive ability, research has largely heralded the benefits of social intelligence as a competence-related attribute that is relevant for formal leadership emergence (e.g., Byrne et al., 2007). This finding may be a direct result of an aptitude for being able to perceive, understand, and regulate one’s emotions (Joseph & Newman, 2010)—emotional competencies that can enhance interactions with subordinates and improve initial impressions with hiring committees within organizations. The benefits of social intelligence are less clear for informal emergent leadership as initial evidence equally supports the contention that being socially competent can be beneficial (e.g., Gruber et al., 2018) or irrelevant to being seen as leaderlike by others (Emery et al., 2011). Holistically, research suggests that social intelligence at minimum does not prevent an individual from becoming a leader, and at maximum can offer opportunities for leadership emergence that otherwise might not be possible without this critical competency.

We note that in addition to the aforementioned antecedents, there are several other promising antecedents that are more distal in nature that have received limited attention to date. These trait antecedents which can be found detailed in Table 2 include

(but are not limited to) concepts such as emotions (i.e., Melwani et al., 2012; Schaumburg & Flynn, 2012), identity (e.g., Harms et al., 2007; Kwok et al., 2018), proactive personality (e.g., Clark & Waldron, 2016; McClean et al., 2018), and dark triad personality traits (e.g., Brunell et al., 2008; Gleason et al., 1978). Further exploration of the distal antecedents included in the text and tables, as well as elaborating on those antecedents that may not have been studied yet, are all reasonable ways to expand knowledge in the leadership emergence domain.

Motivation

As we transition into discussing semidistal needs such as motivational antecedents, we note a relative lack of scholarship compared to other predictors of leader emergence, particularly regarding formal leader emergence (Table 3). In addition, we acknowledge that this was not one of the subcategories that was proposed to be relevant to leadership by DeRue et al. (2011). Nonetheless, pre-existing frameworks such as Mann (1959) acknowledge the role of motivation in these processes. Of the work related to motivation conducted to date, much of the focus has been on the concepts of power and achievement motivation, as well as Chan and Drasgow’s (2001) framework of motivation to lead (MTL). With respect to power, scholars have found in several cases that being motivated to attain power results in formal (e.g., Bark et al., 2016; Schuh et al., 2014) and informal leadership emergence (e.g., Harms et al., 2007). Interestingly, whereas an orientation toward achievement translates favorably toward formal leader emergence (Schuh et al., 2014), it does not impact informal emergent leadership (Sorrentino & Field, 1986). Finally, a general desire or motivation to lead has been connected to becoming a formal leader (i.e., Luria & Berson, 2013; Tharenou, 2001) and informal leader (e.g., Luria & Berson, 2013; Wellman et al., 2019). Below, leveraging the underdevelopment of needs and motivational antecedents of leader emergence, we will discuss areas to expand our understanding of these important drivers of formal and informal leader emergence.

Behaviors

Some of the more proximal antecedents of leadership emergence according to our approach are behaviors that allow one to be recognized as leaderlike (Kanfer, 1992; Mann, 1959; see also Table 4). Notably, our review revealed that besides the well-documented finding that the behavior of participating in group discussions leads to informal leadership emergence (e.g., Bass, 1949; Charlier et al., 2016)—behaviors have not generally been consistently examined for either formal or informal leadership emergence. However, Lanaj and Hollenbeck (2015) made a substantial contribution to our understanding of the relationship between behaviors and emergence by investigating a taxonomy of tangible behaviors—concluding that task, social, and boundary-spanning behaviors each help an individual informally emerge as a leader. Leveraging Lanaj and Hollenbeck’s (2015) definition for each behavioral category, there has been additional support suggesting that task- and social-oriented behaviors can help informal emergence (e.g., Bergman et al., 2014; Wolff et al., 2002)—yet, our understanding of the proximal behaviors giving rise to formal leadership emergence has not progressed beyond a stage of nascency (initial evidence shows the potential benefits of

Table 2
Distal Task Competence and Interpersonal Attribute Antecedents of Formal and Informal Leader Emergence

Antecedents	Informal leadership emergence	Formal leadership emergence
Openness to experience	[+]: Colbert et al., 2012 (.22* to .24*); Côté et al., 2010 (S1: .24*); Hong et al., 2011 (S1: .13*); Kickul & Neuman, 2000 (N.R.); Smith, 2009 (.14* to .21*); Taggar et al., 1999 (.17*); Walter et al., 2012 (S2: .23*); [–]: McClean et al., 2018 (–.17*); [Null]: Balthazard et al., 2009 (–.01); Brunell et al., 2008 (S1: .09); Chiu et al., 2017 (S1: .03 to .02); Coglisier et al., 2012 (.04 to .09); Côté et al., 2010 (S2: .06); Emery et al., 2013 (–.17 to .00); Goldberg, 2003 (–.13 to .08); Harms et al., 2007 (.12); Hu et al., 2019 (S1: –.04; S2: –.08); Kwok et al., 2018 (.06); <i>Lacerenza</i> , 2017 (S1: .47 to .39; S2: –.03 to .04); Lee & Farh, 2019 (S1: –.04 to –.01; S2: .03 to .06); <i>Nahrgang</i> , 2009 (–.10 to .11); Neubert & Taggar, 2004 (.14); Ogunfowora & Bourdage, 2014 (S1: –.07 to .05); Ogunfowora & Bourdage, 2014 (S2: .01 to .12); <i>Park</i> , 2018 (S1: .15); <i>Park</i> , 2018 (S2: –.09); Paunonen et al., 2006 (.01); Prochazka et al., 2018 (–.09); Schauberg & Flynn, 2012 (.15); Spark et al., 2018 (.11); Walter et al., 2012 (S1: –.01); [Other]: Brunell et al., 2008 (S2: .03 to .16*); O'Connor & Jackson, 2010; (.02 to .30*); Won, 2006 (.11 to .74*).	[+]: Byrne et al., 2007 (.14*); Conard, 2020 (.17*); [–]: Wille et al., 2018 (Male: –.01*), [Null]: Furnham et al., 2013 (.06 to .07); Moutafi et al., 2007 (.02); Reichard et al., 2011 (–.02 to .06); Wille et al., 2018 (Female: .01); Wyatt & Silvester, 2018 (–.10); [Other]: Borteyrou et al., 2015 (.17 to .23*)
Conscientiousness	[+]: Balthazard et al., 2009 (.14*); Coglisier et al., 2012 (.15* to .17*); Côté et al., 2010 (S1: .20*; S2: .16*); Gordon, 1952 (.53*); Kwok et al., 2018 (.22*); <i>Lacerenza</i> , 2017 (S2: .37* to .39*); Marinova et al., 2013 (.33* to .42*); Ogunfowora & Bourdage, 2014 (S1: .14* to .29*); .23* to .25*); Paunonen et al., 2006 (.17*); Prochazka et al., 2018 (.18*); Serban et al., 2015 (S1: .21*); <i>Super et al.</i> , 2016 (.25*); Taggar et al., 1999 (.20*), [Null]: Brunell et al., 2008 (S1: .04 to .09); Chiu et al., 2017 (S1: .05 to .10); Colbert et al., 2012 (.10 to .14); Harms et al., 2007 (.06); Hong et al., 2011 (S1: .05); Hu et al., 2019 (S1: .12; S2: .11); Lanaj & Hollenbeck, 2015 (.12); Lee & Farh, 2019 (S1: .05 to .07); McClean et al., 2018 (.00); Melwani et al., 2012 (S3: .11); Neubert & Taggar, 2004 (.04); O'Connor & Jackson, 2010 (–.04 to .11); <i>Park</i> , 2018 (S1: .00; S2: .14); Serban et al., 2015 (S2: .09); Schauberg & Flynn, 2012 (.11); Spark et al., 2018 (.09); Walter et al., 2012 (S1: .13; S2: .06); [Other]: Brunell et al., 2008 (S2: –.12* to .13*); Emery et al., 2013 (.16 to .40*); Goldberg, 2003 (.16 to .28*); <i>Lacerenza</i> , 2017 (S1: .50 to .56*); Lee & Farh, 2019 (S2: .13 to .16*); <i>Nahrgang</i> , 2009 (–.07 to .16*); Smith, 2009 (.03 to .15*); Won, 2006 (.12 to .54*).	[+]: Arvey et al., 2006 (.17*); Furnham et al., 2013 (.10* to .11*); Moutafi et al., 2007 (.13*); Zhang & Arvey, 2009 (.15*). [–]: Wille et al., 2018 (Female: –.03*), [Null]: Atwater et al., 1999 (.12); Byrne et al., 2007 (.11); Conard, 2020 (.09); Reichard et al., 2011 (–.11 to .04); Wille et al., 2018 (Male: –.01); Wyatt & Silvester, 2018 (.13); [Other]: Borteyrou et al., 2015 (.04 to .20*)
Extraversion	[+]: Brunell et al., 2008 (S1: .25* to .30*); Cerny, 2008 (.14* to .15*); Colbert et al., 2012 (.16* to .26*); Côté et al., 2010 (S1: .20*); Holmes et al., 1980 (S1: .38*; S2: N.R.); Hong et al., 2011 (S1: .16*); Kickul & Neuman, 2000 (N.R.); Lee & Farh, 2019 (S1: .16* to .20*; S2: .17* to .19*); Lemoine et al., 2016 (S2: .21*); Melwani et al., 2012 (S3: .21*); Mitchell & Coyle, 2019 (.14*); Mitchell et al., 2021 (S2: .15*); Paunonen et al., 2006 (.20*); Riggio et al., 2003 (N.R.); Serban et al., 2015 (S1: .16*); S2: .24*); Smith, 2009 (.13* to .17*); Spark et al., 2018 (.26*); Taggar et al., 1999 (.22*); Walter et al., 2012 (S1: .25*; S2: .36*); [Null]: Balthazard et al., 2009 (.07); Chiu et al., 2017 (S1: –.05 to .08); Coglisier et al., 2012 (.04 to .08); Côté et al., 2010 (S2: .00); Gordon, 1952 (.09); Harms et al., 2007 (.12); Hu et al., 2019 (S2: –.01 to .09); Hunter & Jordan, 1939 (.06); Kwok et al., 2018 (.15); <i>Lacerenza</i> , 2017 (S1: .21 to .26; S2: .16); Lanaj & Hollenbeck, 2015 (.11); Lemoine et al., 2016 (S1: .04); McClean et al., 2018 (–.04); Neubert & Taggar, 2004 (.31); <i>Park</i> , 2018 (S1: .06; S2: –.01); Prochazka et al., 2018 (.03); Schauberg & Flynn, 2012 (.13); [Other]: Brunell et al., 2008 (S2: .02 to .26*); Emery et al., 2013 (.20 to .38*); Goldberg, 2003 (.10 to .18*); Hu et al., 2019 (S1: .09 to .27*); <i>Middleton</i> , 2018 (.18 to .39*); <i>Nahrgang</i> , 2009 (–.05 to .27*); O'Connor & Jackson, 2010 (.12 to .25*); Ogunfowora & Bourdage, 2014 (S1 and S2: .06 to .27*); Won, 2006 (.04 to .40*); Xu, 2018 (.10 to .32*)	[+]: Arvey et al., 2006 (.23*); Conard, 2020 (.23*); Furnham et al., 2013 (.13* to .14*); Moutafi et al., 2007 (.11*); Wille et al., 2018 (Male: .05*; Female: .05*); Zhang & Arvey, 2009 (.27*). [Null]: Borteyrou et al., 2015 (.09 to .16); Byrne et al., 2007 (.09); Reichard et al., 2011 (.03 to .13); Wyatt & Silvester, 2018 (.00).
Agreeableness	[+]: Balthazard et al., 2009 (.16*); Colbert et al., 2012 (.18* to .19*); Hong et al., 2011 (S1: .12*); Walter et al., 2012 (S1: .28*; S2: .12*); Won, 2006 (.34* to .43*) [–]: McClean et al., 2018 (–.22*), [Null]: Brunell et al., 2008 (S1: .04 to .09); Chiu et al., 2017 (S1: –.03 to –.01); Côté et al., 2010 (S1: .04; S2: .05); Goldberg, 2003 (–.15 to .13); Harms et al., 2007 (.10); Hu et al., 2019 (S1: .09; S2: –.07); Kwok et al., 2018 (.19); <i>Lacerenza</i> , 2017 (S1: –.03 to .17; S2: .16 to .17); Lanaj & Hollenbeck, 2015 (.00); Lee & Farh, 2019 (S1: .00 to .04; S2: .12*);	[Null]: Borteyrou et al., 2015 (.08 to .10); Byrne et al., 2007 (.03); Furnham et al., 2013 (–.01); Moutafi et al., 2007 (.00); Reichard et al., 2011 (.01 to .02); Wille et al., 2018 (Male: .00; Female: .00); Wyatt & Silvester, 2018 (.13). [Other]: Conard, 2020 (–.11 to .15*).

Table 2 (continued)

Antecedents	Informal leadership emergence	Formal leadership emergence
Emotional stability	.01 to .10; Neubert & Taggar, 2004 (–.14); Ogunfowora & Bourdage, 2014 (S1: –.13 to –.04; S2: –.03 to .01); <i>Park, 2018</i> (S1: .00; S2: .10; Paunonen et al., 2006 (–.01); Prochazka et al., 2018 (.16); Schauberg & Flynn, 2012 (–.01); <i>Smith, 2009</i> (.00 to .09); Spark et al., 2018 (.01); Taggar et al., 1999 (–.01); [Other]: Brunell et al., 2008 (S2: –.12* to .12*); Coglisier et al., 2012 (.07 to .14*); Emery et al., 2013 (.06 to .36*); O'Connor & Jackson, 2010 (.06 to .24*) [+]: Balthazard et al., 2009 (.13*); Gordon, 1952 (.22*); Lee & Farh, 2019 (S2: .20* to .22*); Paunonen et al., 2006 (.24*); [Null]: Brunell et al., 2008 (S1: .05 to .07; S2: –.04 to .03); Chiu et al., 2017 (S1: –.03 to .01); Coglisier et al., 2012 (.04 to .08); Colbert et al., 2012 (.14 to .15); Côté et al., 2010 (S1: .13; S2: .08); Emery et al., 2013 (–.01 to .26); Harms et al., 2007 (–.03); Holmes et al., 1980 (S2: N.R.); Hong et al., 2011 (S1: .05); Hu et al., 2019 (S1: .08; S2: –.10); Hunter & Jordan, 1939 (.14); Kwok et al., 2018 (.12); <i>Lacerenza, 2017</i> (S1: .19 to .30; S2: .19 to .21); McClean et al., 2018 (–.00); Neubert & Taggar, 2004 (.04); O'Connor & Jackson, 2010 (.10 to .19); Ogunfowora & Bourdage, 2014 (S1: .01 to .06; S2: –.13 to .01); <i>Park, 2018</i> (S1: .04; S2: –.17); Prochazka et al., 2018 (.08); Schauberg & Flynn, 2012 (.03); Spark et al., 2018 (.05); Taggar et al., 1999 (.10); Walter et al., 2012 (S1: –.08); Walter et al., 2012 (S2: .06); Won, 2006 (–.18 to –.10); [Other]: Goldberg, 2003 (–.05 to .23*); Lee & Farh, 2019 (S1: .07 to .15*); <i>Smith, 2009</i> (–.19* to –.05) [+]: Bergman et al., 2014 (.15* to .20*); <i>Chansler, 2005</i> (.40*); <i>Charlier, 2012</i> (.40*); <i>Dashborough et al., 2007</i> (.20*); <i>D'Innocenzo et al., 2014</i> (.18*); <i>Ferentinos, 1996</i> (.30* to .31*); Gordon, 1952 (.22*); Hollander, 1954 (.30*); Hunter & Jordan, 1939 (.41*); Kickul & Neuman, 2000 (N.R.); Luria et al., 2014 (.23*); (8) <i>Medvin, 2015</i> (.42*); Mitchell & Bommer, 2018 (.18*); Mitchell & Coyle, 2019 (.15*); Reynolds, 1944 (.22*); <i>Roberts, 1995</i> (.21* to .31*); Rubin et al., 2002 (.19*); Rueb et al., 2008 (.07*); Smith & Foti, 1998 (.39* to .43*); Taggar et al., 1999 (.33*); [Null]: Chen et al., 2018 (–.01); Côté et al., 2010 (S2: .07); <i>Edmunds, 1993</i> (.06); Emery et al., 2011 (–.16 to –.22); Hong et al., 2011 (S1: .04); Lim, 2019 (–.01 to .12); Luria & Berson, 2013 (S1: .01); McClean et al., 2018 (–.07); Neubert & Taggar, 2004 (.03); <i>Park, 2018</i> (S2: –.03); Prochazka et al., 2018 (.09); Serban et al., 2015 (S1: .05; S2: .12); Walter et al., 2012 (S1: .06); Walter et al., 2012 (S2: .07); [Other]: Bass et al., 1953 (.17 to .35*); Chiu et al., 2017 (S1: .08 to .28*); Emery et al., 2013 (–.14 to .06*); Foti & Hauenstein, 2007 (.15 to .36*); <i>Gershonoff, 2003</i> (.16 to .24*); Goldberg, 2003 (–.16* to .23*); Kalish & Luria, 2016 (.16 to .24*); <i>Nahrgang, 2009</i> (–.28* to –.08); <i>Packard, 1999</i> (–.07 to .45*); Paunova, 2017 (.00 to .12*); Melwani et al., 2012 (S3: –.05 to .59*); Rychlak, 1963 (.16 to .47*); Wurster & Bass, 1953 (.17 to .32*) [+]: Andrews, 1984 (.38*); Bass et al., 1953 (.29* to .31*); Chen et al., 2018 (.25*); Chiu et al., 2021 (S1: .16*); <i>Karakowsky, 1997</i> (.22* to .33*); <i>Park, 2018</i> (S2: .25*); Paunonen et al., 2006 (.29*); Paunova, 2017 (.38*); Smith & Foti, 1998 (.28* to .31*); Türetgen et al., 2008 (.30* to .34*); [Null]: Chiu et al., 2021 (S3: .04; S4: –.04); Foti & Hauenstein, 2007 (.18); Kalish & Luria, 2016 (.09 to .10); Luria & Berson, 2013 (S1: .13); <i>Medvin, 2015</i> (–.07); <i>Roberts, 1995</i> (.06 to .11); Serban et al., 2015 (S1: .20*); S2: .07); [Other]: Brunell et al., 2008 (S1: .09 to .12*); S2: –.09 to .23*); <i>Ferentinos, 1996</i> (.09 to .19*); <i>Gershonoff, 2003</i> (.14 to .23*); Karakowsky & Siegel, 1999 (.09 to .17*) [+]: Clark & Waldron, 2016 (.29*); Côté et al., 2010 (S2: .20*); Gruber et al., 2018 (S1: .23*); S2: .27*); Walter et al., 2012 (S2: .13*); [Null]: <i>DeTommaso-Kleinert, 2011</i> (.14); <i>Dobosh, 2005</i> (–.04 to .15); Emery et al., 2011 (–.08 to .11); Emery, 2012 (N.R.); Hong et al., 2011 (S2: .06 to .13); Riggio et al., 2003 (N.R.); Shaughnessy et al., 2017 (.26); Walter et al., 2012 (S1: .11); [Other]: Côté et al., 2010 (S1: –.05 to .25*); <i>Dashborough et al., 2007</i> (.00 to .14*); <i>Ferentinos, 1996</i> (.08 to .30*); Hong et al., 2011 (S1: –.12* to .07); Wolff et al., 2002 (.07 to .13*).	[+]: Byrne et al., 2007 (.17*); Furnham et al., 2013 (.10*); Li et al., 2018 (S4: .08*); Moutafi et al., 2007 (.12*); Wille et al., 2018 (Male: .01*); Female: .02*); [Null]: Conard, 2020 (.11); Li et al., 2018 (S2: .04); Reichard et al., 2011 (.00 to .05); Wyatt & Silvester, 2018 (–.04); Zhang & Arvey, 2009 (–.03); [Other]: Borteyrou et al., 2015 (.11 to .23*) [+]: Furnham et al., 2013 (.09* to .10*); Li et al., 2011 (.11* to .15*); Ling et al., 2019 (N.R.); Luria et al., 2014 (.18*); Zhang & Arvey, 2009 (.18*); [Null]: Atwater et al., 1999 (.15); Foti & Hauenstein, 2007 (–.15 to .09); Luria & Berson, 2013 (S2: .09); Siegling et al., 2014 (.12); [Other]: Rosen et al., 1976 (.07 to .17*); Borteyrou et al., 2015 (–.29* to .20*); Reichard et al., 2011 (.14 to .28*) [+]: Atwater et al., 1999 (.14*); Foti & Hauenstein, 2007 (.26*); Li et al., 2011 (.12* to .16*); <i>Zhu & Arvey, 2007</i> (.20*); [Null]: Luria & Berson, 2013 (S2: –.01); <i>Middleton, 2018</i> (–.06 to .09); Tharenou et al., 1994 (.04) [+]: Byrne et al., 2007 (.15* to .24*); Shaughnessy et al., 2017 (.31*); Siegling et al., 2014 (.18*)
Self esteem/self-efficacy/core self-evaluation		
Emotional competence/social skills/political skill		

(table continues)

Table 2 (continued)

Antecedents	Informal leadership emergence	Formal leadership emergence
Other trait-like antecedents	Agency: [+]: Karakowsky & Siegel, 1999 (.42* to .47*); <i>Karakowsky, 1997</i> (.25* to .48*); Kent & Moss, 1994 (.22* to .37*); Kolb, 1997 (.37* to .53*); <i>Kolb, 1999</i> (.22*); Malloy & Janowski, 1992 (.32*); Moss & Kent, 1996 (.30* to .39*); Schock et al., 2019 (S1: .24*), [Null]: Schock et al., 2019 (S2: .19); Turengen et al., 2008 (–.20 to –.13); [Other]: <i>Gershonoff, 2003</i> (.11 to .18*); Aggression [–]: Russell, 1981 (–.13*), [Null]: Limon & LaFrance, 2005 (–.08); <i>Packard, 1999</i> (.02 to .06); Rychlak, 1963 (–.13 to .00); Androgyny: [+]: Turengen et al., 2008 (.34* to .41*), [Other]: Hall et al., 1998 (.08 to .26*); Argumentativeness: [+]: Limon & LaFrance, 2005 (.44*). [Other]: <i>Simmons, 1994</i> (.18 to .31*); Assertiveness: [+]: <i>Chansler, 2005</i> (.16*), [Null]: <i>Lacerenza, 2017</i> (S1: .13 to .28; S2: .24 to .25); Attachment: [+]: Berson et al., 2006 (Secure: .17*); [Null]: <i>Medvin, 2015</i> (Anxiety: .25; Avoidant: –.05); Authoritarian personality: [–]: Hollander, 1954 (–.23*); [Null]: Rychlak, 1963 (–.14 to .04); Communication and listening skills: [+]: Johnson & Bechler, 1998 (.54* to .73*); Mitchell et al., 2021 (S1: .19*; S2: .11*); Communion: [+]: Schock et al., 2019 (S2: .20*). [–]: Malloy & Janowski, 1992 (–.30*); Rueb et al., 2008 (–.07*), [Null]: <i>Gershonoff, 2003</i> (.09 to .12); Kent & Moss, 1994 (–.06 to .08); Kolb, 1997 (–.23 to –.17); Kolb, 1999 (–.11); Schock et al., 2019 (S1: .00); Turengen et al., 2008 (–.17 to –.14); [Other]: Karakowsky & Siegel, 1999 (–.24* to –.12); <i>Karakowsky, 1997</i> (–.22* to –.06); Moss & Kent, 1996 (–.15* to –.12); Competitiveness: [+]: Marinova et al., 2013 (.24*); Creative fluency: [+]: Lee & Farh, 2019 (S1: .11* to .12*), [Null]: Lee & Farh, 2019 (S2: .08 to .10); Cultural intelligence: [+]: Lisak & Erez, 2015 (.16*); Dominance: [+]: Carbonell, 1984 (S2: .38*); Carbonell & Castro, 2008 (.33*); <i>Ferentinos, 1996</i> (.42* to .43*); Fleischer & Chertkoff, 1986 (S1: .34*; S2: .17*); Foti & Hauenstein, 2007 (.34*); Harms et al., 2007 (.16*); Hegstrom & Griffith, 1992 (.47*); Hunter & Jordan, 1939 (.20*); <i>Karakowsky, 1997</i> (.24* to .85*); Lips & Keener, 2007 (.25*); Megargee, 1969 (S1: .28*; S2: .25*); Nyquist & Spence, 1986 (.35*); <i>Roberts, 1995</i> (.32* to .38*); Rueb et al., 2008 (.16*); <i>Rutti, 2009</i> (.21*); Sanchez-Cortes et al., 2011 (.77*); Smith & Foti, 1998 (.19* to .21*); Turengen et al., 2008 (.28* to .36*), [Null]: Carbonell, 1984 (S1: .03); <i>Gershonoff, 2003</i> (.13 to .15); <i>Medvin, 2015</i> (.17). [Other]: <i>Ong, 2015</i> (S3: –.19 to .48*); Rychlak, 1963 (.20 to .31*); Emotions: [+]: Melwani et al., 2012 (S1b: Compassion: .36*; S3: admiration: .31*, compassion: .43*, love: .33*, sensitivity: .19*), [–]: Melwani et al., 2012 (S3: Envy: –.20*), [Null]: Melwani et al., 2012 (S1b: Anger: –.18, love: .22, contempt: .24; S3: Anger: –.07, contempt: –.11, tyranny: .06); Schaumburg & Flynn, 2012 (Shame: –.16 to –.03; Guilt: –.04 to .11); Grit: [Other]: <i>Lacerenza, 2017</i> (S1: .14 to .66*; S2: .12 to .32*); Honesty-humility: [Null]: Ogunfowora & Bourdage, 2014 (S1: –.06 to .03). [Other]: Ogunfowora & Bourdage, 2014 (S2: .04 to .17*); Leader identity: [+]: Harms et al., 2007 (.24*); Kwok et al., 2018 (.48*), [Null]: <i>Xu, 2018</i> (.10 to .15); Locus of control: [Null]: <i>Cerny, 2008</i> (.03 to .06); Machiavellianism: [Other]: Gleason et al., 1978 (inverted U: N.R.); Okanes & Stinson, 1974 (.12 to .25*); Moral disengagement: [–]: Ogunfowora & Bourdage, 2014 (S1: –.23* to –.16*; S2: –.30* to –.21*); Narcissism: [+]: Brunell et al., 2008 (S1: .16* to .25*); Nevicka et al., 2011 (.16*), [Null]: <i>Ong, 2015</i> (S1: .08 to .17; S2: –.19 to .03; S3: –.07 to .20). [Other]: Brunell et al., 2008 (S2: .08 to .30*); Paunonen et al., 2006 (–.06 to .33*); Negative affect: [–]: Spark et al., 2018 (–.33*); Openness to diversity: [+]: Lisak & Erez, 2015 (.16*); Personal values: [Other]: <i>Illies et al., 2005</i> (–.25* to .32*); Physical ability: [+]: Kalish & Luria, 2016 (.38* to .42*); Page, 1935 (.27*), [Null]: McClean et al., 2018 (–.12); Positive affect: [Null]: Spark et al., 2018 (.06); Proactivity/self-leadership: [+]: Clark & Waldron, 2016 (.42*); <i>Super et al., 2016</i> (.17*); [Null]: <i>Chiu et al., 2014</i> (–.02); McClean et al., 2018 (–.08). [Other]: <i>Roberts, 1995</i> (.02 to .18*); Risk taking: [+]: Clark & Waldron, 2016 (.26*); Holmes et al., 1980 (S2: N.R.), [Null]: Holmes et al., 1980 (S1: N.R.); Self-monitoring: [+]: <i>Cerny, 2008</i> (.15* to .20*); Dobbins et al., 1990 (.18* to .20*); Ellis, 1988 (.19*); Ellis & Cronshaw, 1992 (.18* to .27*); Ellis et al., 1988 (.26*); <i>Ferentinos, 1996</i> (.26* to .41*); Hall et al., 1998 (.27* to .40*); Kent & Moss, 1990 (S1: .34*; S2: .19* to .25*);	Affect: [Other]: <i>Zhu & Arvey, 2007</i> (Positive Affect: .24*; Negative Affect: –.08); Dominance: [+]: Conard, (17*); Foti & Hauenstein, 2007 (.25*); Leader identity: [Other]: <i>Middleton, 2018</i> (–.03 to .57*); Machiavellianism: [–]: Conard, 2020 (.18*); Masculinity: [+]: Tharenou, 2001 (.20* to .23*); Personal mastery: [+]: Barling & Weatherhead, 2016 (.11*); Physical ability: [+]: Atwater et al., 1999 (.20*); Task competence/expertise: [+]: Rosen et al., 1976 (.37* to .45*). [Null]: Byrne et al., 2007 (.02); [Other]: Claussen et al., 2014 (.00 to .01*); Self-monitoring: [Null]: Foti & Hauenstein, 2007 (.10); <i>Kilduff et al., 2016</i> (.18)

Table 2 (continued)

Antecedents	Informal leadership emergence	Formal leadership emergence
	<i>Kilduff et al., 2016 (.23*); Rubin et al., 2002 (.12*); Türetgen et al., 2008 (.40* to .50*); Özalp Türetgen et al., 2017 (.27*); Walter et al., 2012 (S2: .22*); [Null]: Côté et al., 2010 (S2: -.04); Dobosh, 2005 (-.06 to .14); Emery et al., 2011 (.01 to .04); Foti & Hauenstein, 2007 (.12); Kwok et al., 2018 (-.08); Medvin, 2015 (-.01); Oddou, 1983 (.13); Roberts, 1995 (.00 to .07); Rueb et al., 2008 (-.02); Rutti, 2009 (-.02); Seites, 1979 (-.04 to .08); Simmons, 1994 (-.06 to .19); Van Der Heide, 2006 (N.R.); Xu, 2018 (.07 to .10); [Other]: Garland & Beard, 1979 (-.18 to .28*); Zaccaro et al., 1991 (.15 to .22*); Self-promotion: [Null]; Lim, 2019 (.02 to .15); [Other]: Lim, 2011 (.10 to .17*); remaining impression management tactics also reported in Lim et al.: -.19* to .22*; Task competence/expertise: [±]; Bunyi & Andrews, 1985 (N.R.); Crockett, 1955 (N.R.); Lacerenza, 2017 (S2: .31*); Stigall, 2005 (.19* to .45*); [Null]; Lacerenza, 2017 (S1: .14); Luria & Berson, 2013 (S1: -.01); Sarker et al., 2009 (.07); Seites, 1979 (.03 to .12); Task/leadership ability: [±]; Charlier et al., 2016 (-.28*); De Souza & Klein, 1995 (.27*); Edmunds, 1993 (.34*); Hall et al., 1998 (.19* to .27*); Stang et al., 1976 (.90*)</i>	

Note. We note the exclusion of several personality taxonomies that do not map onto modern forms of measurement such as Cloninger's temperament and character inventory (O'Connor & Jackson, 2010). Studies were also excluded when they examined personality as a general concept without specifying traits that comprise personality (e.g., Reynolds, 1944). Where appropriate personality traits reported in earlier work were categorized in the above table when they showed clear conceptual overlap with the Big Five traits or the Dark Triad of Personality. Italicized citations reflect unpublished, peer-reviewed studies. * denotes that effects reported in a primary study were statistically significant. N.R. indicates that there was insufficient information to calculate an effect.

transformational leadership and past performance; Bark et al., 2016; Powell & Butterfield, 1994). Given these points, there is a strong need for additional research that provides insight into proximal behaviors that help an individual emerge as a leader. We highlight this as a cornerstone for future research as we consider avenues for future research.

Attributions and Relational Associations

Despite decades of research underscoring the proximal importance that perceptions have on who becomes a leader (DeRue & Ashford, 2010; Lord, 1985; Lord et al., 2001)—perceptual antecedents are almost entirely absent from the formal leadership emergence research domain, and only limited empirical investigative efforts have been undertaken related to informal leader emergence (See Table 5). Initial evidence indicates that both attributions of warmth (i.e., DeRue et al., 2015; Wyatt & Silvester, 2018) and attributions of competence (e.g., informal: DeRue et al., 2015; Wyatt & Silvester, 2018) are beneficial for leader emergence. Peer liking also has been connected to informal leadership emergence, with a majority of studies showing that being liked results in emergent leadership (e.g., Hu et al., 2019; Watson & Hoffman, 2004). Similar to the above conversation related to motivational and behavioral antecedents, perceptual factors offer a rich avenue worthy of additional research in the coming years.

Future Avenues for Research

As is evident from the above summary and the full review (Tables 1–5), there has been extensive investigation into the factors giving rise to the emergence of leaders. However, our review of the literature yielded several fruitful areas that would further our understanding of the leadership emergence phenomenon. First, while significant exploration has been conducted into more distal antecedents of leadership emergence such as gender and personality—less is known about how other stable genetic characteristics and biological factors that can be assessed through neuroscientific and other relevant approaches may influence leadership emergence (cf. Waldman et al., 2011). Such research could explore biological and neurological factors and responses that are relevant to the semidistal motivational and proximal antecedents of leadership emergence. In addition, such research could explore how these factors influence perceptions of prospective followers and the prospective leaders themselves (cf. DeRue et al., 2011).

Regarding the motivational antecedents, a first step could be to begin studying more extensively how the three types of motivation to lead (affective identity, social normative, and noncalculative; Chan & Drasgow, 2001) influence formal and informal leadership emergence. While initial progress has been made related to informal leadership (Table 3), it is unclear whether prospective formal leaders are motivated by a sense of duty (social normative MTL), enjoyment (affective-identity MTL), or altruism (noncalculative MTL)—and the exact way in which each motivation type could impact the emergence process. For this reason, more work is needed to better understand the role of these motives in the leader emergence process. In addition, future work could consider how individuals' motivation to lead might influence their decisions to pursue leadership differently over time as they gain more experience

Table 3*Semidistal Motivational Antecedents of Formal and Informal Leader Emergence*

Antecedents	Informal leadership emergence	Formal leadership emergence
Power motivation	[+]: Harms et al., 2007 (.23*); <i>Park, 2018</i> (S1: .30*); <i>Shaughnessy et al., 2017</i> (.35*); [Null]: <i>Park, 2018</i> (S2: .01); [Other]: Anderson & Wanberg, 1991 (–.10 to .23*).	[+]: Bark et al., 2016 (.21*); <i>Schuh et al., 2014</i> (S2: .37*); <i>Schuh et al., 2014</i> (S3: .41*); <i>Schuh et al., 2014</i> (S4: .29*). [Null]: <i>Shaughnessy et al., 2017</i> (.17)
Achievement motivation	[Null]: Sorrentino & Field, 1986 (N.R.)	[+]: <i>Schuh et al., 2014</i> (S3: .18*); <i>Schuh et al., 2014</i> (S4: .10*)
Motivation to lead	[+]: Brunell et al., 2008 (S1: .44* to .61*); <i>D’Innocenzo et al., 2014</i> (.70*); Luria & Berson, 2013 (S1: .24*); Wellman et al., 2019 (.38*). [Null]: Bass et al., 1953 (–.07 to .00)	[+]: Luria & Berson, 2013 (S2: .28*); Tharenou, 2001 (.12* to .17*)
Other motivational antecedents	Affiliation motivation: [+]: Sorrentino & Field, 1986 (N.R.); General motivation: [+]: Kalish & Luria, 2016 (.21* to .28*); Individual commitment to group goal: [+]: De Souza & Klein, 1995 (.16*); <i>Oddou, 1983</i> (.56*); Leadership enjoyment: [+]: Kolb, 1997 (.43* to .45*); Kolb, 1999 (.33*); Learning goal orientation: [+]: Mitchell & Coyle, 2019 (.20*); MTL-affective identity: [+]: Hong et al., 2011 (S1: .24*); Hong et al., 2011 (S2: .21*); MTL-socio-normative: [+]: Hong et al., 2011 (S2: .27*); [Null]: Hong et al., 2011 (S1: .09); MTL-non-calculative: [+]: Hong et al., 2011 (S1: .20*); [Null]: Hong et al., 2011 (S2: .07); Need for status: [+]: <i>Park, 2018</i> (S1: .61*); [Null]: <i>Park, 2018</i> (S2: .22); Prosocial motivation: [+]: Mitchell & Bommer, 2018 (.22*). [Null]: Hu et al., 2019 (S1: .13; S2: –.04); Romance of leadership: [Null]: Cerny, 2008 (.00 to .08); Work engagement: [+]: <i>Bullington, 2016</i> (.49*); <i>Joe, 2016</i> (.33*)	

Note. Italicized citations reflect unpublished, peer-reviewed studies. * denotes that effects reported in a primary study were statistically significant. N.R. indicates that there was insufficient information to calculate an effect.

leading, more deeply identify as a leader, and understand more clearly the costs associated with leading.

Beyond motivation to lead, research should leverage additional theories to identify other plausible motivational predictors of formal and informal leadership emergence. For instance, Hogan’s (1982) socioanalytic theory positions the motives of “getting ahead” versus “getting along” as core motivational drivers. On the surface, one may surmise that those motivated to get along may be stymied from leadership roles because they do not necessarily take charge in leadership settings whereas those motivated to get ahead may disproportionately emerge as formal and informal leaders—potentially to the detriment of a team. Likewise, regulatory focus theory indicates that employees are motivated to behave based on their approach or avoidance orientation (Elliot & Thrash, 2002). An employee who is approach oriented may enthusiastically rise to the challenge of leadership—whereas an employee who is oriented toward avoidance may consider the risks or costs of leadership too great to actively pursue the role of formal or informal leader (Zhang et al., 2020). Research on these and other theory-driven motivational antecedents has the potential to make an important contribution to our understanding of what it is that encourages prospective formal and informal leaders to persist toward emergence in the face of adversity or challenging conditions.

We also surmise that less empirical attention has been given to the behaviors driving leadership emergence as there is not a robust taxonomy of behaviors that precede being a leader. That is, once an individual becomes a leader they can be seen as engaging in a host of behaviors such as being a transformational, transactional,

considerate, or task-oriented leader to name a few. Yet, the focus of each construct is a repertoire of behaviors that typically are conducted when an individual already occupies a leadership role. Thus, to advance theory on the behavioral antecedents of leader emergence—survey instruments should be developed or refined to account for the fact that individuals are often not yet in leadership roles. Additional scholarship in this area will offer practical value as scholars and practitioners work toward helping employees identify tangible behaviors they can engage in to emerge as both formal and informal leaders.

Beyond the above antecedents, we also encourage future research to focus on the role that attributions and perceptions have in the emergence of formal and informal leadership emergence and how they might mediate other antecedents outlined above and depicted in Figure 1. As noted earlier in this review, the ability to emerge as a leader is largely predicated on attributions and perceptions—yet recent research in this area has tended to remain focused toward theoretical pursuits and qualitative endeavors (cf. DeRue et al., 2015). Concerted focus toward understanding how these factors driving leadership emergence mediate other antecedents may shed light on why demographic minorities may be hindered in emerging as leaders, why prototypes people possess of leaders result in specific personality types occupying leadership roles more frequently (e.g., extraverts vs. introverts), and why some individuals struggle to emerge as a leader despite possessing several key indicators of emergence (e.g., an extraverted individual who is perceived as less receptive because they stifle a group’s proactivity; Grant et al., 2011). Other areas of worthwhile inquiry may involve

Table 4
Proximal Behavioral Antecedents of Formal and Informal Leader Emergence

Antecedents	Informal leadership emergence	Formal leadership emergence
Participatory behavior (quantity and quality)	[+]: <i>Adams, 2009</i> (.46* to .57*); <i>Balthazard et al., 2009</i> (.15* to .45*); <i>Bass, 1949</i> (.93*); <i>Benjamin, 1995</i> (N.R.); <i>Bergman et al., 2014</i> (.29* to .66*); <i>Burke, 1974</i> (.81*); <i>Charlier et al., 2016</i> (.31* to .48*); <i>Crockett, 1955</i> (N.R.); <i>Huang, 2010</i> (.39*); <i>Jaffee & Lucas, 1969</i> (.63*); <i>Jiang et al., 2015</i> (.65*); <i>Jones & Kelly, 2007</i> (N.R.); <i>Karakowsky, 1997</i> (.22* to .96*); <i>Kirscht et al., 1959</i> (.54*); <i>Lana et al., 1960</i> (N.R.); <i>Li et al., 2007</i> (.25*); <i>Malloy & Janowski, 1992</i> (N.R.); <i>Morris & Hackman, 1969</i> (.46* to .65*); <i>Oddou, 1983</i> (.84*); <i>Riggio et al., 2003</i> (N.R.); <i>Sanchez-Cortes et al., 2013</i> (.46* to .51*); <i>Seties, 1979</i> (.22* to .76*); <i>Sorrentino & Field, 1986</i> (.52* to .69*); <i>Stang et al., 1976</i> (.89*); <i>Ward, 1968</i> (N.R.); <i>Watson & Hoffman, 2004</i> (.88*); <i>Wentworth & Anderson, 1984</i> (.37* to .62*); [Other]: <i>Anderson & Wanberg, 1991</i> (.07 to .42*); <i>Burroughs & Jaffee, 1969</i> (.13 to .45*)	[Other]: <i>Rosen et al., 1976</i> (.11 to .15*)
Transformational leadership	[+]: <i>Gershenoff, 2003</i> (.36* to .48*); <i>Prochazka et al., 2018</i> (.97*); [Other]: <i>Ong, 2015</i> (SI: .25 to .85*); <i>Ong, 2015</i> (S2: .09 to .54*)	[+]: <i>Bark et al., 2016</i> (.18*); <i>Li et al., 2012</i> (.29*); [other]: <i>Reichard et al., 2011</i> (.17 to .21*)
Task-oriented behaviors	[+]: <i>Benjamin, 1995</i> (N.R.); <i>Bergman et al., 2014</i> (.28* to .70*); <i>Berkowitz, 1956</i> (.52*); <i>Gershenoff, 2003</i> (.37* to .56*); <i>Hite, 2009</i> (.74*); <i>Lanaj & Hollenbeck, 2015</i> (.72*); <i>Mitchell et al., 2019</i> (.33*); <i>Mitchell & Bommer, 2018</i> (.19*); <i>Oddou, 1983</i> (.24*); <i>Park, 2018</i> (S2: .70*); <i>Trask, 1999</i> (.56*); <i>Walter et al., 2012</i> (SI: .53*); <i>Walter et al., 2012</i> (S2: .59*); <i>Wolff et al., 2002</i> (.14*); [Null]: <i>Schneier & Bartol, 1980</i> (N.R.); [Other]: <i>Foti & Hauenstein, 2007</i> (.12 to .56*); <i>Nahrgang, 2009</i> (.13 to .72*)	[+]: <i>Weinberg et al., 2019</i> (.23*); [Null]: <i>Foti & Hauenstein, 2007</i> (.01 to .15)
Social-oriented behaviors	[+]: <i>Benjamin, 1995</i> (N.R.); <i>Bergman et al., 2014</i> (.28* to .57*); <i>Hite, 2009</i> (.64*); <i>Lanaj & Hollenbeck, 2015</i> (.43*); <i>Park, 2018</i> (S2: .60*); <i>Schneier & Bartol, 1980</i> (N.R.); <i>Xu, 2018</i> (16* to .63*); <i>Wolff et al., 2002</i> (.11*), [Null]: <i>Oddou, 1983</i> (–.14); [Other]: <i>Mitchell et al., 2019</i> (Curvilinear $\gamma = -0.12^*$; $r = .35^*$); <i>Nahrgang, 2009</i> (.00 to .45*)	[+]: <i>Weinberg et al., 2019</i> (.18*)
Boundary-spanning behaviors	[+]: <i>Lanaj & Hollenbeck, 2015</i> (.46*); <i>Liu et al., 2018</i> (.23*); [Other]: <i>Nahrgang, 2009</i> (.10 to .42*)	[+]: <i>Deshpande et al., 1994</i> (N.R.); <i>Kilduff et al., 2016</i> (.25*); <i>Powell & Butterfield, 1994</i> (.10*); <i>Yap & Konrad, 2009</i> (N.R.); [Null]: <i>Breaugh, 2011</i> (–.04); <i>Opper et al., 2015</i> (N.R.)
Past performance	[+]: <i>Kilduff et al., 2016</i> (.35*); <i>Zhang et al., 2012</i> (.13*); [Null]: <i>Lim, 2019</i> (–.07 to .04)	
Other behavioral antecedents	Change participation: [+]: <i>Gerpott et al., 2019</i> (.21* to .34*); Communication apprehension: [–]: <i>Charlier et al., 2016</i> (–.33*); <i>Limon & LaFrance, 2005</i> (–.43*); [Other]: <i>Dobosh, 2005</i> (–.09 to .08); <i>Simmons, 1994</i> (–.22 to –.36*); Facilitative leadership: [+]: <i>Guastello, 1995</i> (N.R.); Helping: [+]: <i>Marinova et al., 2013</i> (.31*); Nonverbal: [+]: <i>Jiang et al., 2015</i> (.71*); [Other]: <i>Baird, 1977</i> (–.15 to .45*); <i>Bhattacharya, 2019</i> (–.25* to .49*); <i>Gerpott et al., 2018</i> ; (N.R.); Perspective taking: [–]: <i>Super et al., 2016</i> (–.21*); Proactive behavior: [+]: <i>Zhang, 2018</i> (.19*); Social participation: [+]: <i>Charlier, 2012</i> (.22*); <i>Huang, 2010</i> (.23*); [Null]: <i>Adams, 2009</i> (.05); <i>Hawkins, 1995</i> (–.07); [Other]: <i>Gerpott et al., 2019</i> ; (TI: .13 to .35*); Task participation: [+]: <i>Adams, 2009</i> (.55*); <i>Gerpott et al., 2019</i> (.27* to .43*); <i>Hawkins, 1995</i> (.67*); <i>Huang, 2010</i> (.29*); Teamwork behaviors: [+]: <i>Luria & Berson, 2013</i> (SI: .44*); Voice: [+]: <i>McClean et al., 2018</i> (.31* to .38*)	Negative behavior (Sustained complaints): [–]: <i>Bishopp, 2013</i> (N.R.); Positive behavior (Commendations): [Null]: <i>Bishopp, 2013</i> (N.R.)

Note. Italicized citations reflect unpublished, peer-reviewed studies. * denotes that effects reported in a primary study were statistically significant. N.R. indicates that there was insufficient information to calculate an effect.

Table 5*Proximal Attribution and Relational Antecedents of Formal and Informal Leader Emergence*

Antecedents	Informal leadership emergence	Formal leadership emergence
Warmth	[+]: DeRue et al., 2015 (.46* to .55*); Zhang, 2018 (.25*)	[+]: Wyatt & Silvester, 2018 (.90*)
Competence	[+]: DeRue et al., 2015 (.74* to .87*); Ho et al., 2012 (.52*); Rubin et al., 2002 (.12*); Sanchez-Cortes et al., 2011 (.30*); Watson & Hoffman, 2004 (.72*); Zhang, 2018 (.36*)	[+]: Wyatt & Silvester, 2018 (.70*)
Liking	[+]: Hu et al., 2019 (S1: .39*; S2: .19*); Lacerenza, 2017 (S2: .38* to .58*); Oddou, 1983 (.54*); Stang et al., 1976 (.74*); Trask, 1999 (.45*); Watson & Hoffman, 2004 (.59*); [–]: Sanchez-Cortes et al., 2011 (–.30*); [Other]: Lacerenza, 2017 (S1: .35 to .86*); Seites, 1979 (.20 to .39*)	
Other perceptual antecedents	Charisma: [+]: Cohen & Yoon, 2021 (.12* to .61*); Leader potential: [+]: Benjamin, 1995 (N.R.); Perceived influence/power: [+]: Harms et al., 2007 (.27* to .39*); Oddou, 1983 (.90*); Trask, 1999 (.76*); Watson & Hoffman, 2004 (.57*); [Other]: Park, 2018 (S2: .02 to .61*); Perceived status/prestige: [+]: McClean et al., 2018 (.42*); [Other]: Ong, 2015 (.12 to .67*); Park, 2018 (S2: –.01 to .42*); Perceived trust: [+]: Charlier, 2012 (.46*); Marinova et al., 2013 (.31*); Prototypicality: [+]: Roberts, 1995 (.23* to .59*); Rutti, 2009 (.31*); [Null]: Berson et al., 2006 (–.05 to .11); [Other]: Nahrgang, 2009 (.00 to .58*)	Leader potential: [+]: Atwater et al., 1999 (.15*); Breaugh, 2011 (.34*); Powell & Butterfield, 1994 (.19*); Rosen et al., 1976 (.24* to .29*); Prototypicality: [Null]: Schuh et al., 2014 (S2: .07)

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investigating how prospective leaders successfully or unsuccessfully manage the perceptions others possess through strategies such as self-promotion and ingratiation (Bolino & Turnley, 2003), and how to overcome an initial impression that one is not well suited for the role of a leader. Collectively, novel endeavors related to understanding attributions and perceptual antecedents should also shed important insight into how other people factor into one's leadership ascendancy.

Though not depicted in our model, a final area that may offer fruitful inquiry is examining the recursive impact that being a leader has on certain antecedents that give rise to being a leader. This focus has been largely absent from the literature—yet, important questions exist related to how being a leader impacts one's traits, motives, and behaviors—and how changes in these various domains could subsequently impact future leadership emergence. For instance, being a leader could both increase and decrease an individual's motivation to lead depending on whether they are enjoying the role itself or see it as rewarding. Likewise, one may not initially embrace a leader identity—however, they may come to possess that role identity over time due to exposure and experience as a leader. Finally, while we typically think of personality as stable, scholars have noted that personality has the potential to evolve over time due to important life events (Specht et al., 2011). Becoming a leader may reflect such an event in one's life and the experience could influence aspects of one's personality, such as conscientiousness or agreeableness, because they are now fully responsible for their team or because they have to make difficult decisions that may not always be well received by subordinates. Leadership might also influence aspects of personality such as narcissism and humility that have been looked at as antecedents. Unpacking the trajectories of these aspects of personality and how such trajectories relate to new opportunities to emerge (or not) as a leader in other contexts would be useful. Altogether, the potential recursive effects between the antecedents of leadership emergence and the actual experience of emerging as a leader is an area of great promise moving forward.

Question 3: What Outcomes Are Associated With Leader Emergence?

As one considers the nomological network of leader emergence, natural questions arise related to the potential outcomes associated with individuals emerging as formal and informal leaders. To date, studies focusing on the consequences of leadership emergence are limited in number. Where such work has been done, there has often only been a single study looking at each of the relationships, and there are ample opportunities for follow-up research that replicates or extends these initial findings (Table 6). Notwithstanding, initial evidence has shown that informal leaders can influence team dynamics, as well as performance. In particular, informal leaders strengthen social support among members (Berson et al., 2006; Crozier et al., 2017) and the presence of informal leadership in team settings has more often than not contributed to effective team performance (e.g., De Souza & Klein, 1995; Wellman et al., 2019; Zhang et al., 2012). Not surprisingly, the individual outcomes of being an emergent informal leader oneself are generally positive, with informal leaders being more satisfied at work (e.g., Özalp Türetgen et al., 2017), experiencing a higher degree of organizational commitment (e.g., Lim, 2011), and being seen as more effective leaders/performers (e.g., Berson et al., 2006; Colbert et al., 2012; Lanaj & Hollenbeck, 2015)—though we would like to remind readers to interpret these findings with caution as they are based on a limited number of studies.

Even less research has explored the consequences for the individual of being a formal leader—yet, initial evidence demonstrates that formal leadership emergence can result in higher levels of satisfaction (Barling & Weatherhead, 2016) and commitment at work (Bagdadli et al., 2006). Moreover, individuals who had emerged as formal leaders were found to have higher salaries (e.g., Barling & Weatherhead, 2016; Weinberg et al., 2019) and they generally found the workplace to be more procedurally just than nonleaders (Bagdadli et al., 2006). Finally, across several studies, Li et al. (2018) concluded that individuals who had emerged as formal leaders had a higher degree of well-being (relative to nonleaders).

Table 6
Outcomes of Formal and Informal Leader Emergence

Consequences	Informal leadership emergence	Formal leadership emergence
Task performance/ contribution to team	[+]: <i>Chiu et al., 2014</i> (.40*); <i>Cogliser et al., 2012</i> (.30* to .63*); <i>Colbert et al., 2012</i> (.63*); <i>Joe, 2016</i> (.17*); <i>Kalish & Luria, 2016</i> (.51* to .53*); <i>Lee & Farh, 2019</i> (S2: .50* to .72*); <i>Liu et al., 2018</i> (.14*); <i>Oddou, 1983</i> (.42*); <i>Sarker et al., 2009</i> (.37*); <i>Shaughnessy et al., 2017</i> (.31*). [Null]: <i>Foti & Hauenstein, 2007</i> (.08 to .11); <i>Nevicka et al., 2011</i> (–.01). [Other]: <i>Lee & Farh, 2019</i> (S1: –.08 to .29*); <i>Xu, 2018</i> (–.07 to .28*)	[Null]: <i>Foti & Hauenstein, 2007</i> (Job performance: .02, Overall performance: .05); <i>Shaughnessy et al., 2017</i> (.22)
Leader effectiveness	[+]: <i>Berson et al., 2006</i> (.37*); <i>Hite, 2009</i> (.72*); <i>Lacerenza, 2017</i> (S1: .78* to .94*); <i>Lacerenza, 2017</i> (S2: .88* to .93*); <i>Lanaj & Hollenbeck, 2015</i> (.77*); <i>Luria et al., 2014</i> (.18* to .36*); <i>Luria et al., 2019</i> (.07* to .11*); <i>Prochazka et al., 2018</i> (.82*)	[+]: <i>Atwater et al., 1999</i> (.52*); [Other]: <i>Wyatt & Silvester, 2018</i> (.04 to .25*)
Follower/team performance	[+]: <i>Borg, 1957</i> (N.R.); <i>De Souza & Klein, 1995</i> (.23*); <i>Joe, 2016</i> (.47*); <i>Prochazka et al., 2018</i> (.40*); <i>Wellman et al., 2019</i> (.47*); <i>Zhang et al., 2012</i> (.20*). [Null]: <i>Cogliser et al., 2012</i> (.01 to .11). [Other]: <i>Karakowsky, 1997</i> (–.12 to .37*)	
Job/team satisfaction	[+]: <i>Joe, 2016</i> (.40*); <i>Lawson, 2016</i> (.10*); <i>Özalp Türetgen et al., 2017</i> (.36*). [Null]: <i>Chiu et al., 2021</i> (S1: –.09); <i>Chiu et al., 2021</i> (S3: –.07); <i>Chiu et al., 2021</i> (S4: –.12); <i>Super et al., 2016</i> (–.12). [Other]: <i>Cook et al., 2019</i> (.11 to .37*)	[+]: <i>Barling & Weatherhead, 2016</i> (.10*); [Null]: <i>Campisi, 2002</i> (.17)
Other consequences of emergent leadership	Creative performance: [Null]: <i>Illies et al., 2005</i> (–.13 to .12). [Other]: <i>Xu, 2018</i> (–.05 to .18*); Counterproductive behavior: [–]: <i>Hite, 2009</i> (–.29*); Democratic behaviors toward followers: [Null]: <i>Crozier et al., 2017</i> (.05); Intent to leave: [–]: <i>Joe, 2016</i> (–.31*); Meaning schema transformation: [+]: <i>Chen et al., 2018</i> (.69*); Organizational commitment: [+]: <i>Lim, 2011</i> (.19* to .28*); [Other]: <i>Bullington, 2016</i> (.00 to .25*); <i>Özalp Türetgen et al., 2017</i> (.08 to .30*); Positive feedback for followers: [Null]: <i>Crozier et al., 2017</i> (.13); Social support for followers: [+]: <i>Berson et al., 2006</i> (.37*); <i>Crozier et al., 2017</i> (.20*); Talk time in group: [+]: <i>Leeming, 2019</i> (N.R.); Team trustworthiness: [Other]: <i>Cogliser et al., 2012</i> (–.09 to .22*); Training for followers: [+]: <i>Crozier et al., 2017</i> (.20*)	Blood pressure: [Null]: <i>Li et al., 2018</i> (S2: –.04 to .07). [Other]: <i>Li et al., 2018</i> (S3: –.10* to –.08); Chronic diseases: [Null]: <i>Li et al., 2018</i> (S2: –.03; S4: –.06); Democratic behaviors toward followers: [+]: <i>Crozier et al., 2017</i> (.20*); Environmental behaviors: [+]: <i>Salvador & Burciaga, 2020</i> (.09* to .10*); Informal leader emergence: [+]: <i>Foti & Hauenstein, 2007</i> (.29*); <i>Guo et al., 2021</i> (.27*); <i>Luria et al., 2014</i> (.36*); <i>O'Rourke, 1999</i> (.28*); <i>Shaughnessy et al., 2017</i> (.87*). [Other]: <i>Packard, 1999</i> (–.02 to .49*); Intent to leave: [Null]: <i>Bagdadli et al., 2006</i> (–.13); Organizational commitment: [+]: <i>Bagdadli et al., 2006</i> (.21*); Participation in decision/policy making: [+]: <i>Keels, 1996</i> (.46* to .52*); Positive feedback for followers: [Null]: <i>Crozier et al., 2017</i> (.09); Procedural justice: [+]: <i>Bagdadli et al., 2006</i> (.28*); Salary: [+]: <i>Asiamah & Mensah, 2019</i> (.22*); <i>Barling & Weatherhead, 2016</i> (.40*); <i>Keels, 1996</i> (.20*); <i>Weinberg et al., 2019</i> (.24*); <i>Yap & Konrad, 2009</i> (N.R.); <i>Zhang & Arvey, 2009</i> (.23*); Salivary cortisol: [–]: <i>Li et al., 2018</i> (S3: –.16*). [Null]: <i>Li et al., 2018</i> (S2: –.02; S4: .11); Social support for followers: [+]: <i>Crozier et al., 2017</i> (.34*); Training for followers: [+]: <i>Crozier et al., 2017</i> (.41*); Well-being: [+]: <i>Li et al., 2018</i> (S2: .07*; S4: .13* to .19*); Whistleblowing intentions: [Other]: <i>Asiamah & Mensah, 2019</i> (.05 to .25*)

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Future Research Avenues

Altogether, the relative lack of research looking at outcomes of emergent leadership leaves open the door for an important area of inquiry. Researchers might explore both the personal benefits and drawbacks of ascending into formal and informal leadership roles. *Zhang et al.'s (2020)* recent work highlights the risks associated with leadership—such as elevated responsibility and blame, costs to personal relationships, and threats to one's image—yet it is unclear if and how these risks translate to negative workplace outcomes once an individual has emerged as a leader. Accordingly, evaluating

whether (and when) informal/formal leaders are blamed for failure, experience a decrease in friendship ties when becoming a leader, or are viewed in a less positive light by others are all potential starting points to better understand negative outcomes of leadership ascendance. Future research might also delve into how emerging as a leader might result in an inflated self-view or even increasingly narcissistic orientation that might undermine relationships and one's ability to be successful in the leadership role. Despite the negative personal outcomes emergent leaders may experience, personal benefits can also arise when one becomes a formal or informal leader. Some avenues for inquiry include whether informal and formal leaders experience

better career prospects through the increased access to information and resources afforded through being powerful, having higher status, and occupying more central positions in social networks within organizations. Similarly, emergent leaders may be in a position to influence policy, practices, and decision-making in an organization that offers personal benefits in both the short and long terms. These, as well as other prospective personal benefits for emergent leaders, should be at the forefront of research moving forward.

Beyond personal benefits, increased understanding of the impact that leader emergence has on team outcomes is imperative. While evidence indicates that the presence of an informal leader can aid team performance, it is unclear through what emergent, transition, and interpersonal processes this occurs (Marks et al., 2001). For instance, does the emergence of informal leadership in a team enhance goal specification prior to a task being carried out or serve as a key motivational factor inspiring members to persist in the face of challenges? Questions such as these may enhance understanding of the process through which informal leaders contribute to team performance. As alluded to in the previous point, research to date almost exclusively indicates that informal leaders benefit teams. It is clear however that leaders come in a variety of forms, some of which imbue toxicity into teams such as narcissistic individuals who may be exploitative and entitled or abusive leaders who bully and ridicule those around them (cf. Galvin et al., 2015). In light of these maladaptive attributes, more work is needed on when informal leadership is negative. This research could also be extended to formal leadership to understand when an individual's designation as team leader might result in process loss or other negative repercussions. Altogether, we encourage future researchers to begin exploring when informal and formal leader emergence might translate to negative outcomes for those they lead.

A final area of future research is related to the concepts of over- and underemergence (Lanaj & Hollenbeck, 2015). Research has largely viewed emergent and effective leadership in separate silos—yet, Lanaj and Hollenbeck (2015) highlighted the intricate relationship between the two, mainly that the wrong people occasionally emerge into leadership roles. To tease out these relationships, they viewed informal leader emergence in tandem with effectiveness—noting that overemergers become leaders more often than their effectiveness ratings would dictate and underemergers become leaders less often than their effectiveness ratings would dictate. Despite this fascinating assertion, research has subsequently provided few additional insights into the consequences that being an “emergent leader” has on being an “effective leader.” Accordingly, enhanced understanding regarding both the personal and team consequences that being an over- or underemerged leader have is warranted. That is, are teams with over- or underemerged leaders less effective or experiencing more harmful team dynamics relative to teams where leaders with commensurate emergence and effectiveness ratings occupy informal and formal positions of leadership? It would also be important for researchers to unpack how overemergence influences the psychology of the followers and leaders themselves, and how those who were passed over approach their careers and roles differently after observing overemergence. The answers to these questions are largely unknown, yet integral for enhancing team dynamics and maximizing team performance.

Question 4: What Are the Boundary Conditions of Leadership Emergence?

Beyond understanding the antecedents, outcomes, and phenomenon of leadership emergence, it is important to understand the boundary conditions that act as moderators of these relationships (see Tables 7 and 8). Some research has begun to explicitly explore the complexity of leadership emergence by looking at how boundary conditions such as how the presence or absence of certain factors, or the degree of a certain variable, might alter relationships and outcomes relevant to leadership emergence. This research also contributes to our overall understanding of leadership emergence's nomological network.

The most common approach has involved looking at the moderating influence of individual attributes (Table 8). For instance, research has looked at core self-evaluation as a potential boundary condition. Paunova (2017) found that in elite international teams, status related to development of home nation and English proficiency were tied to informal leader emergence. However, these relationships were moderated by core self-evaluations of the prospective leader. Specifically, she found that core self-evaluations strengthen a somewhat weak effect of English proficiency on informal leadership emergence, and that core self-evaluations help to compensate for lower levels of one's home country's development (Paunova, 2017). In addition, intelligence has been looked at as a moderator that is relevant to leadership emergence. Luria and Berson (2013) found that intelligence of the prospective leader interacts with motivation to lead to predict informal leader emergence, such that the relationship between motivation to lead and leader emergence is stronger for individuals higher in intelligence.

Research exploring the boundary conditions relevant to leadership emergence has also looked at how group or organizational factors influence emergence. Liu et al. (2018) found that the relationship between employee boundary-spanning behavior and informal leadership emergence is enhanced in the context of high-power distance. Research also suggests that the effectiveness of a group's trainer in leadership development programs will strengthen the relationship between informal leadership emergence and eventual effectiveness as a leader (Luria et al., 2019).

Interestingly, research has also explored how the situation itself may influence who emerges as leaders. This research relies on the assumption that individuals emerge as leaders in various situations in part because the capabilities or traits of the individual are well suited to the task at hand or specific needs of the organization (Barnlund, 1962). This research stream also suggests that individuals' ability to perceive a situation and respond accordingly might interact with one's traits to allow them to meet the needs of the situation (Kenny & Zaccaro, 1983; Zaccaro et al., 1991).

Finally, research has also considered measurement features that can impact the findings of a study. For instance, Zaccaro et al. (1991) found that self-monitoring predicted emergent leadership when group members ranked peers (from top leadership choice to last leadership choice), but not when participants reported Likert leadership ratings for peers using the general leadership impression scale. Moss and Kent (1996) concluded that agentic and communal traits more strongly predicted self-reported emergent leadership ratings as compared to other-reported emergent leadership ratings. Finally, Emery et al. (2011) demonstrated that emergent leadership

Table 7
Contextual Factors Relevant to Formal and Informal Leader Emergence

Structural considerations		Informal leadership emergence	Formal leadership emergence
Team attributes	Network characteristics	Need for informal leadership: [+]: Wellman et al., 2019 (.45*); Perceived team's competence: [+]: DeRue et al., 2015 (.16*); Perceived social support: [+]: Joe, 2016 (.39*); <i>Task</i>, 1999 (.26*); Xu, 2018 (.19* to .64*); Zhang, 2018 (.17*); [Null]: Chiu et al., 2021 (S1: .11; S4: .00); [Other]: Duncan, 2011 (–.11 to .55*); Perceived team's warmth: [+]: DeRue et al., 2015 (.23* to .24*); Prior experience with team members: [+]: DeRue et al., 2015 (.06* to .08*); [Null]: Hu et al., 2019 (S1: .04); Lim, 2019 (–.03 to .07); Prior team performance: [+]: Chiu et al., 2021 (S1: .15*); [Null]: Hu et al., 2019 (S2: .01); Psychological safety: [+]: McClean et al., 2018 (.23*); Team ethnic diversity: [Null]: Chiu et al., 2021 (S3: –.08); Lee & Farh, 2019 (S2: –.07 to –.03); [Other]: Lee & Farh, 2019 (S1: –.12* to –.08); Team friendship: [Other]: Cook et al., 2019 (.08 to .16*); Team gender composition: [Null]: Chiu et al., 2021 (S3: –.04); Lim, 2019 (–.03 to .05); Neubert & Taggar, 2004 (.05); Team identification: [+]: DeRue et al., 2015 (.29* to .37*); <i>Hite</i>, 2009 (.49*); Team initiative climate: [+]: Zhang, 2018 (.20*); Team Leader-Member Exchange (LMX) variability: [Null]: Zhang et al., 2012 (.04); Team major diversity: [Null]: Lee & Farh, 2019 (S1: .00 to .05; S2: –.01 to –.02); Team member contributions: [Other]: Lee & Farh, 2019 (S1: –.19* to .11*; S2: –.09 to .23*); Team power distance: [Null]: Liu et al., 2018 (.11); Team task interdependence: [Null]: Neubert & Taggar, 2004 (–.13); Team size: [–]: Chiu et al., 2021 (S3: –.17*); Cook et al., 2019 (–.37*); Mitchell et al., 2019 (–.09*), [Null]: Balthazard et al., 2009 (.05); <i>Bhattacharya</i>, 2019 (–.04); Chiu et al., 2021 (S1: –.08); Colbert et al., 2012 (–.13); <i>D'Innocenzo et al.</i>, 2014 (.07); Lee & Farh, 2019 (S1: .07 to .10; S2: –.02 to .03); Lim, 2019 (–.13 to .08); Mitchell & Bommer, 2018 (–.13); Mitchell & Coyle, 2019 (.02); Wellman et al., 2019 (.03); Zhang, 2018 (–.12); Zhang et al., 2012 (–.01); Team tenure: [+]: Chiu et al., 2021 (S1: .17*); [Null]: Zhang et al., 2012 (.04); Virtual team: [Null]: Adams, 2009 (–.06); Serban et al., 2015 (S1: .00; S2: –.02); [Other]: Cook et al., 2019 (.10 to .60*); Lim, 2011 (–.31* to .02) Advice brokerage: [Other]: Xu, 2018 (–.04; .31*); Advice centrality: [+]: Chiu et al., 2017 (S1: .15* to .22*); Hu et al., 2019 (S1: .43*; S2: .24*); <i>Kilduff et al.</i>, 2016 (.74*); Kwok et al., 2018 (.59*); Neubert & Taggar, 2004 (.57*); [Null]: Chiu et al., 2017 (S2: .23); Betweenness centrality: [+]: Kwok et al., 2018 (.48*); Sutaranto et al., 2011 (N.R.); Boundary-spanning ties: [Other]: Guo et al., 2021; (.14 to .18*); Closeness centrality: [Null]: Sutaranto et al., 2011 (N.R.); Communication network centrality: [+]: Goldberg, 1955 (.42*); Cyclical ties: [+]: Emery et al., 2011 (N.R.); Degree centrality: [+]: Bullington, 2016 (.26* to .31*); Sutaranto et al., 2011 (N.R.) [Null]: <i>Dashborough et al.</i>, 2007 (.08); Eigenvector centrality: [+]: Bullington, 2016 (.34*); Friendship brokerage: [Other]: Xu, 2018 (–.11 to .25*); Negative in-degree centrality: [–]: Chiu et al., 2017 (S2: –.30*); [Null]: Chiu et al., 2017 (S1: –.08); Outdegree centrality: [–]: Emery et al., 2011 (N.R.); Preferred friendship: [+]: Berkowitz, 1956 (.35*); Prior friendships: [+]: Emery et al., 2011 (N.R.); Lana et al., 1960 (N.R.); Reciprocity: [+]: Emery et al., 2011 (N.R.); Social status/popularity: [+]: Emery et al., 2011 (N.R.); Paunova, 2017 (.81*); Stang et al., 1976 (.84*); [Null]: Li et al., 2007 (–.04); Tertius iungens: [Null]: Xu, 2018 (–.04 to .11); Tie density: [–]: Bullington, 2016 (–.39* to –.21*); [Null]: Serban et al., 2015 (S1: .00; S2: .02); Transitivity: [+]: Emery et al., 2011 (N.R.); Trust brokerage: [+]: Kilduff et al., 2016 (.36*)	Team Size: [–]: Claussen et al., 2014 (–.12* to –.03*) Advice centrality: [+]: Kilduff et al., 2016 (.60*); Homophily: [+]: Oppen et al. (2015; N.R.); Internal hire: [+]: Powell & Butterfield, 1994 (.25*); Network size: [–]: Claussen et al., 2014 (–.06* to –.02*); Organizational connections: [+]: Deshpande et al., 1994 (N.R.); Trust brokerage: [Null]: Kilduff et al., 2016 (.08)

Table 7 (continued)

Structural considerations	Informal leadership emergence	Formal leadership emergence
Job and situational factors	Job demands: [+]: Clark & Waldron, 2016 (.16*)	Career encouragement: [+]: Tharenou, 2001 (.08* to .07*), [Null]: Tharenou et al., 1994 (.01); Career support: [+]: Tharenou, 2001 (.09* to .11*); Cognitive job requirements: [Null]: Maume, 1999 (N.R.); Educational encouragement: [Null]: Tharenou et al., 1994 (.05); Hours worked: [+]: Maume, 1999 (N.R.); Job control: [+]: Li et al., 2018 (S1: .12* to .24*; S2: .31*; S3: .21*; S4: .34*); Job demands/challenging job: [+]: Li et al., 2018 (S1: .05 to .13*; S2: .21*; S3: .23*; S4: .21*); Tharenou, 2001 (.27*); Percent Black in occupation: [–]: Maume, 1999 (Female: N.R.), [Null]: Maume, 1999 (Male: N.R.); Percent female in occupation: [+]: Maume, 1999 (Male: N.R.), [–]: Maume, 1999 (Female: N.R.); Training/developmental opportunities: [+]: Arvey et al., 2007 (.35*); Tharenou, 2001 (.52* to .44*); Tharenou et al., 1994 (.36*); [Other]: Fassiotto et al., 2018 (N.R.); <i>Middleton, 2018</i> (–.03 to .30*); Sector: [Other]: Tharenou, 2001 (.02 to .09*); Industry: [Other]: Maume, 1999 (N.R.); Nelson et al., 2019 (N.R.); Male hierarchy of organization: [Null]: Tharenou, 2001 (–.03 to .02); Tharenou et al., 1994 (–.04); Organization size: [Null]: Tharenou, 2001 (–.04 to .00); Organizational stressors: [Other]: <i>Duncan, 2011</i> (–.49* to .21*); Union presence: [–]: Maume, 1999 (N.R.)
Organizational/industry characteristics	Industry type: [Null]: Chiu et al., 2017 (S2: –.02); Responsiveness of partner organization: [+]: Chen et al., 2018 (.25*)	
External leader attributes	Formal leader charisma: [+]: Wellman et al., 2019 (.27*); Formal leader laissez faire: [–]: Wellman et al., 2019 (–.25*); Formal leader's effective leadership: [+]: Wellman et al., 2019 (.48*); Leader member exchange: [+]: Zhang et al., 2012 (.13*)	
Other contextual factors	Comfort with technology: [Null]: Serban et al., 2015 (S1: .09; S2: .14); Least preferred coworker: [–]: Schmeier, 1978 (–.21*); [Null]: Rice & Chermers, 1973 (N.R.); Seating arrangement: [+]: Howells & Becker, 1962 (.27*); Ward, 1968 (N.R.)	

Note. Italicized citations reflect unpublished, peer-reviewed studies. * denotes that effects reported in a primary study were statistically significant. N.R. indicates that there was insufficient information to calculate an effect.

Table 8*Hypothesized Exemplar Boundary Conditions Impacting Formal and Informal Leader Emergence*

Moderators examined	Informal leadership emergence	Formal leadership emergence
Measurement/experiment features	Measure type: Zaccaro et al., 1991; Self-other report: Anderson & Wanberg, 1991; Colbert et al., 2012; Moss & Kent, 1996; Repeated measurement of leader emergence: Emery et al., 2011; Spillman et al., 1981; Sex of role model: Geis et al., 1985; Task type: Garland & Beard, 1979; Gleason et al., 1978; Hall et al., 1998; Ho et al., 2012; O'Connor & Jackson, 2010; Wentworth & Anderson, 1984	
Team attributes	Company representative's responsiveness: Chen et al., 2018; Competence shift in team: Cook et al., 2019; Density of network ties: Serban et al., 2015; Gender diversity of Team: Bunyi & Andrews, 1985; Gurman & Long, 1992; Karakowsky & Siegel, 1999; Lemoine et al., 2016; Group extraversion: Lemoine et al., 2016; Group goals: De Souza & Klein, 1995; Group power distance: Liu et al., 2018; Peer contributions: Lee & Farh, 2019; Presence of a designated leader: Borg, 1957; Reward interdependence: Nevicka et al., 2011; Team cohesion: Sarker et al., 2009; Team leadership: Taggar et al., 1999; Team shared vision: Zhang et al., 2012; Team type: Elgar, 2016; Serban et al., 2015; Virtual vs. face to face/amount of face to face contact: Balthazard et al., 2009; Cook et al., 2019	
Personal attributes	Agency × Communion: Schock et al., 2019; Age × Genetics: Chaturvedi et al., 2012; Cognitive ability: Luria & Berson, 2013; Emotion recognition × Extraversion: Walter et al., 2012; English proficiency × Core self-evaluation: Paunova, 2017; Facets of extraversion × Prosocial motivation: Hu et al., 2019; Gender × Big Five traits: Neubert & Taggar, 2004; Gender × Dominance: Carbonell, 1984; Hegstrom & Griffith, 1992; Megargee, 1969; Gender × Dominance × Feedback: Fleischer & Chertkoff, 1986; Gender × Dominance × Incentives: Lips & Keener, 2007; Gender × Dominance × Role model: Carbonell & Castro, 2008; Gender × Gaze: Gerpott et al., 2018; Gender × Genetics: Chaturvedi et al., 2012; Gender × Information: Watson & Hoffman, 2004; Gender × Self-monitoring: Dobbins et al., 1990; Ellis, 1988; Gender × Task type × Self-Monitoring: Ellis & Cronshaw, 1992; Gender × Voice: McClean et al., 2018; Intelligence × Dominance × General self-efficacy: Foti & Hauenstein, 2007; Smith & Foti, 1998; Motivation to lead: Wellman et al., 2019; Political skill: Shaughnessy et al., 2017	Gender: Barling & Weatherhead, 2016; Li et al., 2011; Maume, 1999; Melamed & Bozionelos, 1992; Tharenou, 2001; Wille et al., 2018; Gender × Dependent Children: Windsor & Auyeung, 2006; Gender × Race: Yap & Konrad, 2009
Other moderators	Trainer's leadership effectiveness: Luria et al., 2019	Environmental orientation × Managerial status: Salvador & Burciaga, 2020; Job type: Rosen et al., 1976; Social environment: Zhang et al., 2009

ratings were impacted by the time of measurement, whereby the positive association between an individual's self-identity as a leader and their leadership emergence ratings become stronger across time. In summary, this suggests that how and when we choose to measure leader emergence, as well as who reports ratings of leadership emergence each can have important implications regarding the conclusion drawn in a given study—and accordingly should be carefully considered when designing studies of emergent leadership.

Future Avenues for Research

Because of the limited amount of research looking at boundary conditions related to leader emergence, there remains substantial opportunities to further unpack what we currently know. Our results revealed several broad categories of more contextually oriented

factors that have been considered as control variables or antecedents of leader emergence including team attributes (e.g., identification with team: DeRue et al., 2015; team size: Claussen et al., 2014; Lee & Farh, 2019), network characteristics (e.g., advice centrality; Chiu et al., 2017; Hu et al., 2019), task or situational factors (e.g., job demands; Clark & Waldron, 2016; Li et al., 2018), organizational/industry characteristics (e.g., industry type; Maume, 1999), and external leader attributes (e.g., Wellman et al., 2019; Zhang et al., 2012; see also Table 7). Despite the recognition of the possible role these factors have in emergent leadership—this research has remained siloed from the person-oriented antecedents discussed earlier in this review. As such, we call upon researchers to adopt an interactionist perspective (House et al., 1996) that more deeply considers the interplay between person-oriented antecedents and features of the context.

To help prompt future research in this domain, we offer a few high-level examples below. We hope to underscore to readers the relative nascency of this area of leadership emergence research—and highlight the considerable necessity for future scholarship to account for contextual factors moving forward. First, research would benefit from understanding how both the attributes of a team and the context within which that team operates influence the relationships outlined in our distal–proximal framework of leadership emergence. Of particular note, structural considerations of a team, such as how many members it is comprised of, how long each member has been involved with the team, and a team's current hierarchical structure can each offer nuanced moderating implications for whether an individual can emerge into a leadership position. Likewise, the level of interdependence among members, the type of team (e.g., action, project, top management), and whether individuals are on multiple teams are worthy considerations when trying to determine who will emerge as leaders—as each likely creates a unique context that lends itself to the emergence of certain types of leaders (e.g., introverted leaders, female leaders). We underscore the necessity of examining the broader organizational context, including factors such as organizational culture, policies, degree of focus on inclusion, and the political landscape within the company. Of particular emphasis, researchers may focus on the impact that employee-fit within an organization has in facilitating (or impeding) formal and informal leader emergence (Kristof-Brown et al., 2005). Factors such as these may help to explain when certain individuals, such as underrepresented minorities, less tenured employees, or other persons who do not align with conventional leader prototypes are more or less likely to emerge into leadership positions.

Beyond the context within a team and organization, it might also be worthwhile to add clarity regarding environmental and societal factors that can impact leadership emergence within organizations. Factors range drastically and might include environmental dynamism, health pandemics, natural disasters, and acts of terrorism. A common thread through these categories external to an organization is that each change the level of uncertainty and resource availability, creating the possibility of economic turbulence and social unrest. Historically, political leaders who have been seen as extremely toxic such as Adolf Hitler and Robert Mugabe have capitalized on uncertainty to catapult themselves first into a position of informal leadership, and later into a position of formal leadership. It is plausible that a similar pattern may occur for leadership within organizations—where abhorrent or toxic leaders emerge into leadership roles due to uncertainty external to the organization—a feat that would otherwise not be possible during times of normalcy (see relevant research on the toxic triangle; Padilla et al., 2007).

Finally, we underscore a broad necessity to examine the moderating influence that time has in determining one's leadership emergence. Only recently have scholars begun to explore the dynamic nature of leadership emergence (Gerpott et al., 2019; Lee & Farh, 2019), yet scholars have frequently echoed the criticality of temporal considerations when studying organizational phenomenon (George & Jones, 2000; Shipp & Cole, 2015). In this regard, it is possible that leadership emergence, particularly in an informal context, is largely fluid and subject to change based on the team's current tasks. For instance, in a research and development team—informal leadership at one stage of the project may be fulfilled by an engineer who is responsible for the initial design

of a product and transition to a manufacturing member when the team transitions to product creation. Future scholars may also consider incorporating time-related constructs such as time pressure and time management (Clark & Waldron, 2016; Shipp & Cole, 2015)—as these concepts may alter the necessity for leaders to emerge in a team or organizational setting. It would also be interesting to consider time as a moderator related to understanding the success or failure of leaders, and explore how some leaders might emerge too early or too late in the evolution of their team's development or their own personal development. Altogether, we encourage researchers to account for the “essential impact of context” (Johns, 2006, p. 386) that has long been recognized as important in organizational behavior through the consideration of team, organizational, external, and temporal factors.

Discussion

Theoretical Implications

Our conceptual review of the formal and informal leadership emergence literature highlights both areas of maturity, as well as nascency upon which future research can be developed. Somewhat uniquely, our review broadly leverages research from multiple disciplines including (but not limited to) business, communications, criminal justice, education, healthcare, and psychology. In so doing, we contribute to several areas of theory including leadership, individual differences, and teams.

First, this review makes an important contribution to leadership research. In particular, although several meta-analyses exist that contribute to our understanding of leader emergence, most have been focused on a single or small subset of antecedents (e.g., intelligence, narcissism, the Big Five traits)—with little coverage of the process giving rise to leader emergence (cf. Badura et al., 2018, 2020), and even less attention has been given to the outcomes of emergent leadership. Relatedly, qualitative syntheses have tended to concentrate on antecedents and mediators of leader emergence while largely overlooking the moderators and consequences of leadership emergence (Acton et al., 2019; Paunova, 2015; cf. Hanna et al., 2021). Finally, even the most comprehensive reviews of leader emergence to date have been focused on informal forms of leadership, leaving questions about the phenomenon of formal leader emergence and its relationship with informal leader emergence (e.g., Hanna et al., 2021). Our review seeks to address this lack of attention to formal leadership emergence by beginning the process of bridging these two types of leadership emergence to understand the complex process of leadership emergence and the nuanced differences that occur in the nomological network of formal and informal leadership emergence.

Second, our research advances theory on individual differences in organizational settings by clarifying the sequential ordering of a broad array of antecedents that have been studied in the leader emergence literature. Since Kanfer's (1990, 1992) introduction of the distal–proximal framework of motivation—it has often been used to argue for the sequential ordering of a small set of variables (e.g., Christian et al., 2009; Ng et al., 2008; Stewart et al., 2012) and only recently have scholars used it to organize a more expansive set of antecedents in the leadership literature (Badura et al., 2020). Accordingly, in this review, we leverage Kanfer's (1990, 1992) approach and DeRue et al.'s (2011) categorizations to organize

attributes based on their causal proximity to emergent leadership. By doing so, we underscore the merits of studying the *process* of leader emergence and the need for more clarity related to potential mediational processes, instead of opting to only examine a small subset of antecedents without consideration of nuanced mediational effects—a general trend that has unfolded in the past several decades of research on emergent leadership. Altogether, we hope to catalyze leadership emergence research that better accounts for the holistic attributes of prospective leaders and the process of emergence itself rather than the emphasis on a single attribute which may under-represent an individuals' potential as an emergent leader.

Third, our research has broad implications for research on teams. In particular, despite the relative lack of studies examining leader emergence consequences—our review helps make clear the potential benefits that emergent leaders have on group processes and performance. That is, emergent leaders—whether formal or informal—occupy an important role in teams whereby they can help usher in positive performance through adaptive group processes such as offering social support for followers (Berson et al., 2006; Crozier et al., 2017), helping teams form a high degree of trust (Cogliser et al., 2012), and training subordinates to be more effective in their role (Crozier et al., 2017). An additional implication of our review on the team's literature is that the somewhat rosy outlook of emergent leaders that has permeated research on team consequences thus far may be contingent on certain adaptive characteristics of the leader and environment. Accordingly, our identification of a set of moderating factors that may accentuate the darker side of organizational behavior—mainly having a toxic individual emerge into a formal or informal leadership position—serves as an interesting development for team-level research that is worthy of further empirical and theoretical exploration. We note that a logical next step following this review would be to work toward further integrating the leadership emergence literature at the individual level of analysis with that at the team level (e.g., shared leadership, distributed leadership). While outside the purview of this qualitative synthesis, we underscore the necessity of recognizing that emergent leadership does not occur within a vacuum and that others may vie for leadership positions meaning that multiple individuals could occupy informal or formal leadership. By leveraging the work reported in this review, and considering it in the context of teams where multiple individuals may emerge as leaders, we may further enhance understanding of when emergent leadership may contribute favorably or detrimentally to team outcomes.

Practical Implications

Beyond implications for research, our integrative review provides important insight to guide practitioners. Among these practical insights is the identification of the factors giving rise to informal and formal leadership emergence. As noted in our introduction, leadership development is a multibillion-dollar industry, yet this costly investment for organizations may yield only marginal returns if it is invested in individuals who do not have the temperament or motivation to emerge into and persist in leadership positions. By pinpointing factors that predispose one to emerge as a leader and shedding light on the process of emergence itself, we provide practitioners with additional understanding of the leadership phenomenon that will help them facilitate the emergence and success of

those who will best contribute to the goals and well-being of the organization.

In addition, through the process of introducing a distal–proximal framework of leadership emergence—we highlighted a set of attributes and behaviors that may be malleable and others that are more fixed in nature. This understanding has the potential to help organizations better channel development efforts for those seeking leadership positions. In particular, the most malleable predictor of leadership emergence is behaviors—thus by highlighting recent work identifying task, social, and boundary-spanning behaviors as key antecedents of leadership emergence, we provide insight for trainers on how to best teach individuals to be leaderlike. For instance, organizations can focus on tailoring development initiatives to teach individuals how to improve task, boundary-spanning, and social behaviors such as “coordinat[ing] the work done by others,” “go[ing] outside the team to bring in new resources,” and “listen[ing] carefully to the thoughts and feelings of others” (Lanaj & Hollenbeck, 2015; pp. 1482–1483). Furthermore, by identifying the distal antecedents that drive task, social, and boundary-spanning behaviors (e.g., personality, motives)—trainers may identify a prospective leader's comfort or preference toward engaging in certain behaviors and work toward capitalizing behavioral strengths and minimizing weaknesses to help improve odds of emergent leadership.

Finally, we underscore the potential influence that contextual factors may have in altering one's ascendance into a formal or informal position of leadership. Generalizability of leadership emergence raises a critical concern for practitioners, as research has frequently been studied without accounting for the differences that may occur across situations. Thus, certain demographics, personality types, and behaviors that give rise to leadership in one organization may not facilitate a similar rise to leadership in another. Likewise, the personal and team outcomes of leadership emergence may not always yield positive outcomes as has often been the case in research to date. For practice, our review serves as an important reminder for the relative need for caution when broadly consuming and applying knowledge received from scientific investigations in a complex field such as leadership. Consequently, if an organization is working to create a rich pipeline of future leaders, accounting for the uniqueness of their organization will be beneficial to increase the likelihood of informal and formal leader emergence among their high potential individuals.

Conclusion

In the introduction of this manuscript, we underscored the theoretical importance and practical emphasis organizations place on leadership and the emergence of formal and informal leaders. Through our review of this vast and complex literature, we found that despite the preconceived notion that leadership emergence research has reached a stage of relative maturity, there remains considerable room for additional scientific inquiry and theoretical development for both the study of formal and informal emergent leadership. By integrating the existing work—we highlight both under- and well-established aspects of the nomological network of leadership emergence—with the ultimate goal of building a platform that can further enhance our understanding of this important aspect of the leadership phenomenon for years to come.

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